2026 prelim mechanics questions

Mechanics questions from the 2026 prelim directory.
Whole pages are kept. A page that also holds another question is included in full.
RI items are the marked scripts. There is no blank RI question paper in the corpus.

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Answer **all** the questions in the spaces provided.

1 **(a)** A ball of mass 0.80 kg is released from rest in air from a tall building. The variation of the velocity of ball v with time t is shown in Fig. 1.1.

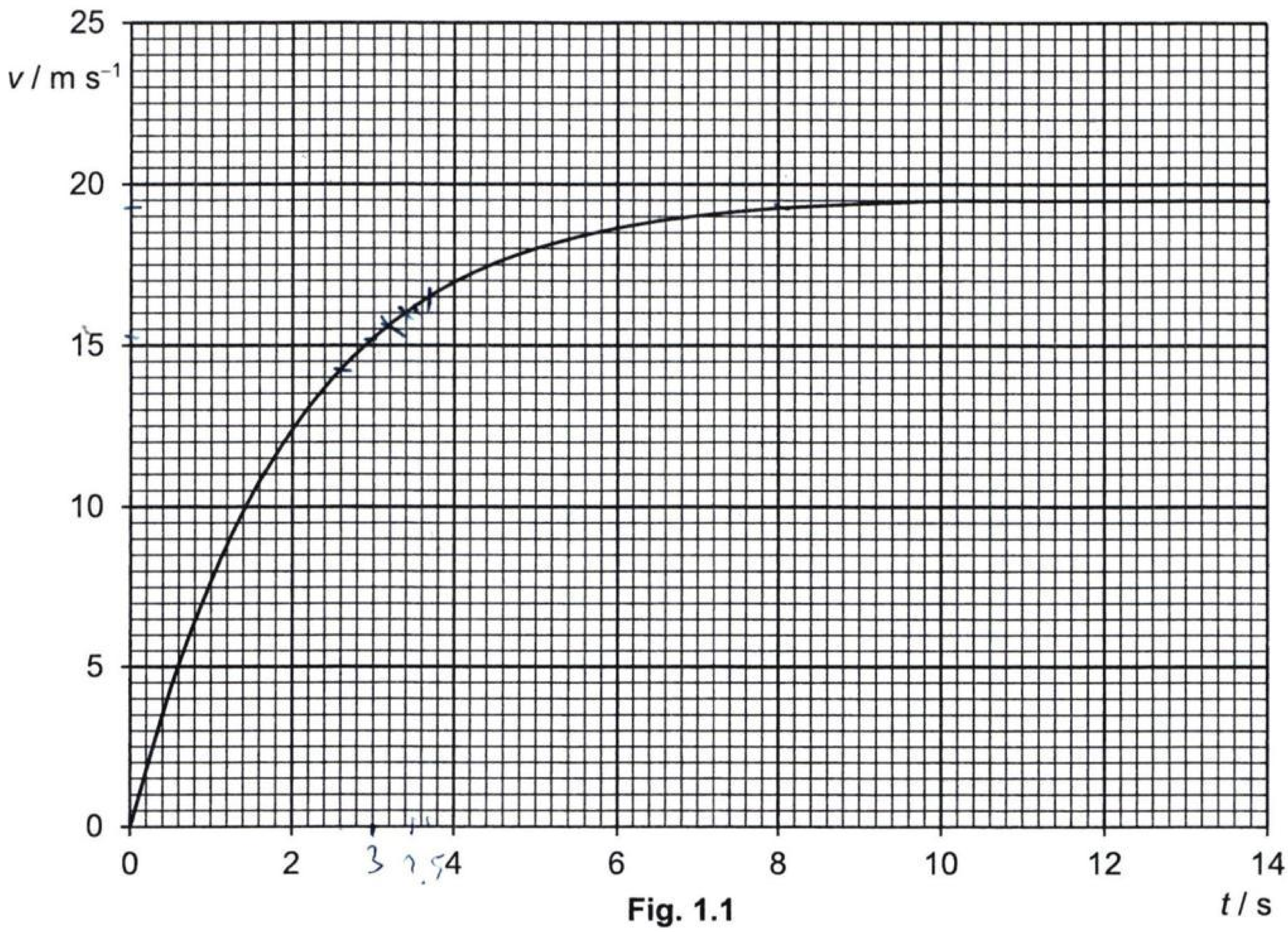


Fig. 1.1

(i) With reference to Fig. 1.1, explain how the drag force F_D acting on the ball changes as the velocity of ball increases from t = 0 to t = 8.0 s.

1 / 2

The drag force is directly proportional to the velocity, ~~in the~~ opposing the direction of velocity. By newton's second law, F=ma and the force in opposite direction has an effect on the velocity such that the velocity ~~de~~ increases at a decreasing rate, ~~suggesting that the~~ and faces a ~~greater~~ flatter gradient as the velocity increases toward 19.5m s^{-1}, suggesting that the opposing force increases proportionate to velocity. [2]

X

(ii) Determine the magnitude of the drag force F_D acting on the ball at time t = 3.5 s.

1 / 3

at $t = 3.5$s, $V =$ ~~15.25 ms^{-1}~~ $16.25\ ms^{-1}$

~~At~~ $m = 8.0kg$ $F_w = (8.0)(9.81)$

~~At~~ ~~8.98~~

using (3.4, 16.0),

$F_{net} = (8.0)(2.5)$

$\text{gradient} = \frac{16.25-16.0}{3.5-3.4}$

$= 2.5$ ~~x~~ $1.7\ ms^{-1}$

$F_D = F_w - F_{net}$

$= (8.0)(9.81-2.5)$

$= 58.48$

$= 58N$ (2 s.f.)

~~At~~ gradient $\frac{dv}{dt} = a$

$F_D =$ ..... 58 ..... ~~x~~ N [3]

(b) The drag force F_D is given by the equation

$$F_D = 6\pi r \eta v$$

where r is the radius of the ball,
η is the viscosity of air,
v is the velocity of the ball.

(i) Determine the SI base units of η.

3 / 3

$F_D = 6\pi r \eta v$

$kg\ m\ s^{-2} = 6\pi\ m^2\ s^{-1}\ \eta$

$6\pi\eta = kg\ m^{-1}\ s^{-1}$

base SI units is $kg\ m^{-1}\ s^{-1}$

base units = ... ~~kg m^{-1}~~ $kg\ m^{-1}\ s^{-1}$ ..... [3]

(ii) Explain how the maximum drag force will change if a ball of a greater mass, but the same radius, is released from rest from the same building.

0 / 1

The maximum drag force does not change as drag force at max is unrelated to mass. ~~It is dependent on~~

..... [1]

[Total: 9]

Section A

Answer **all** the questions in this Section in the spaces provided.

1 Two spheres A and B of masses 69 g and 250 g respectively are suspended by light inextensible threads from fixed points. When the spheres are at rest, they are in contact with their centres at the same horizontal level, as shown in Fig. 1.1.

The vertical distance from the fixed point of suspension to the centre of sphere A is 1.50 m.

A force F, acting at an angle θ to the vertical, is applied to sphere A, displacing it so that the thread makes an angle of 24° with the vertical, as shown in Fig. 1.2.

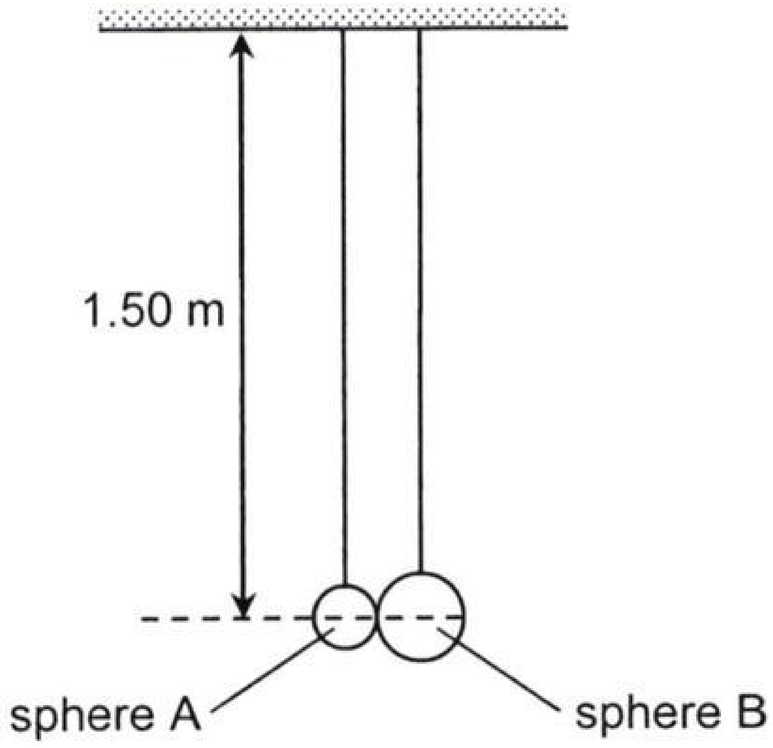


Fig. 1.1 (not to scale)

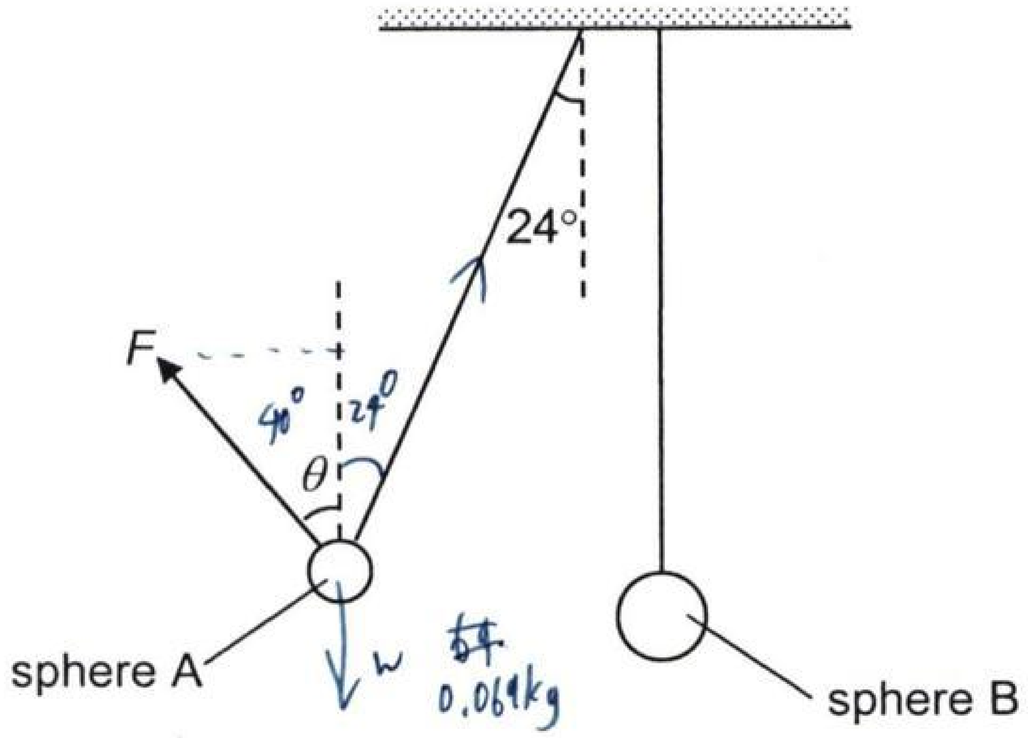
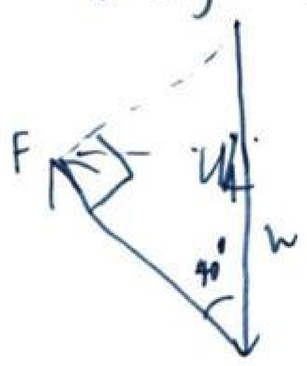


Fig. 1.2 (not to scale)

(a) (i) Calculate the magnitude of the force F required to hold sphere A in equilibrium when θ is 40°.

0 / 3

weight of A = (0.069)(9.81) = 0.6769



F = (0.6769) cos(40°)
= 0.5185
= 0.52 N (2 s.f.)



F = 0.52 N [3]

(ii) By considering the vector triangle for the forces acting on sphere A, deduce the value of θ for which the force F required to maintain equilibrium is a minimum.

0 / 1

at 90°, smallest F

θ = 90 ° [1]

(b) The force F is removed and sphere A is released from rest. It subsequently collides with sphere B.

(i) Determine the work done by the gravitational force on sphere A, from the point of release to the instant just before collision.

0 / 1

work done by the gravitational force = ... J [1] × (0)

(ii) Hence, show that the momentum of sphere A just before collision is 0.11 kg m s^{-1}.

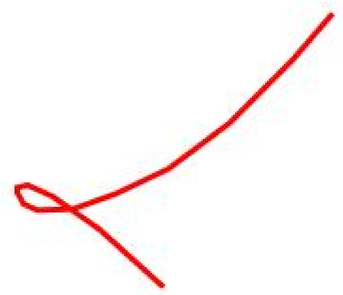
0 / 1 W =

[1] × (0)

(c) After the collision, sphere A rebounds in the opposite direction while retaining 52% of its kinetic energy just before impact. The remainder of its kinetic energy is either transferred to sphere B or transformed into other forms of energy.

(i) Calculate the momentum of sphere B after the collision.

0 / 3



momentum = ... kg m s^{-1} [3]

(ii) A student states that if the collision were such that sphere A retains none of its initial kinetic energy, more kinetic energy would be transferred to sphere B.

0 / 2

Comment on the validity of this statement.

.....

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.....

.....

..... [2]

× (0)

[Total: 11]

2 The radius of the Earth is 6.4×10^{6} m and the mass of the Earth is 6.0×10^{24} kg.

A spacecraft of mass 1200 kg is in a circular orbit at an altitude of 400 km above Earth's surface.

(a) Calculate the orbital speed of the spacecraft.

2 / 2

gravitational accel = centripetal accel

$r = 6.4\times10^{6} + 400\times10^{3} = 6.8\times10^{6}$ m

$$\frac{GM}{r^2} = \frac{v^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$= \sqrt{\frac{(6.67\times10^{-11})(6.0\times10^{24})}{6.8\times10^{6}}}$$

$= 7672$ m

$= 7700$ m s^{-1} (2 s.f.)

orbital speed = 7700 m s^{-1} [2]

(b) Calculate the gravitational potential energy of the spacecraft.

2 / 2

$$G.P.E = -\frac{GMm}{r}$$

$$= -\frac{(6.67\times10^{-11})(6.0\times10^{24})(1200)}{6.8\times10^{6}}$$

$= -7.062\times10^{10}$

$= -7.1\times10^{10}$ J (2 s.f.)

gravitational potential energy = −7.1 ×10^{10} J [2]

(c) Determine the additional energy required for the spacecraft to just reach infinity.

Explain your working.

2 / 3

Spacecraft in its orbit currently has $v^2 = \frac{GM}{r}$ from (a).

$$K.E. = \frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{r}$$

Energy required to reach infinity is $-\frac{GMm}{r}$ from (b).

M to reach infinity, spacecraft still needs $\left|-\frac{GMm}{r} + \frac{1}{2}\frac{GMm}{r}\right| = \frac{GMm}{2r}$ J of energy

additional energy $= \frac{1}{2}\left(-\frac{GMm}{r}\right)(-1)$

$= \frac{7.062\times10^{10}}{2}$ from (b).

$= 3.531\times10^{10}$

$= 3.5\times10^{10}$ J (2 s.f.)

energy = 3.5 ×10^{10} J [3]

(d) The total energy for the spacecraft in orbit at any orbital radius r is given by

$$-\frac{GMm}{2r}$$

where M is the mass of Earth and m is the mass of spacecraft.

The spacecraft is now moved to a circular orbit closer to Earth.

(i) State the changes in the kinetic energy and gravitational potential energy of the spacecraft.

0 / 1

kinetic energy will increase but gravitational potential energy will increase at twice the increase of kinetic energy

[1]

(ii) State and explain whether positive or negative work must be done on the spacecraft.

2 / 2

Negative work done. The total amount of energy decreases and negative work is done to do so.

[2]

[Total: 10]

3 A trolley of mass m is held on a smooth horizontal surface by means of two identical springs, each of force constant k and unstretched length L. The springs are attached to fixed points as shown in Fig. 3.1.

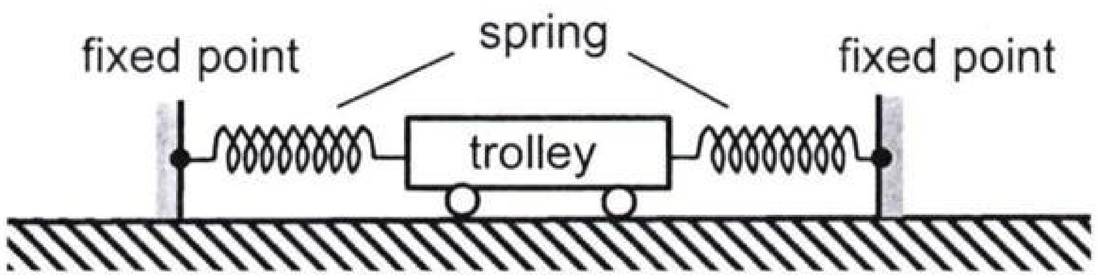


Fig. 3.1

(a) When the trolley is in equilibrium, the extension of each spring is e.

(i) The trolley is displaced horizontally to the right and then released at time $t = 0$. The trolley then oscillates with simple harmonic motion. The springs remain stretched throughout the oscillation.

0 / 2

Show that the acceleration a of the trolley is given by

$$a = -\frac{2k}{m}x$$

where x is the displacement of the trolley from its equilibrium position.

Explain your working.

by newton's second law, $F = ma$

hooke's law, $F = -kx$

~~both springs experience compression and springback by turn~~

both springs stretch and bounce

$ma = -kx$ (2) ✓

$a = -\frac{2k}{m}x$ (shown)

[2]

1 An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m s^{-1} as shown in Fig. 1.1.

A is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

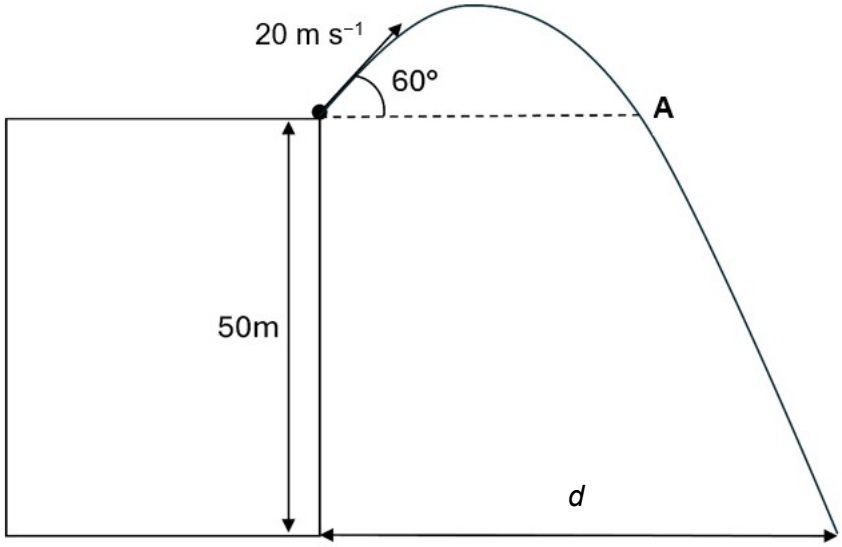


Fig. 1.1

(a) Assuming there is no resistance.

(i) Calculate the distance *d*.

d = ... m [3]

(ii) Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

……..

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…… [2]

(b) Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

(i) Show quantitatively that the air resistance is not negligible in this case.

[2]

(ii) The speed of the object at its maximum height is 8.0 m s^{-1}.
Calculate the energy lost due to air resistance at maximum height.

energy lost = ... J [2]

(c) Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]

2 In a collision between two bodies A and B, the force that A exerts on B varies with time in the way shown in Fig. 2.1.

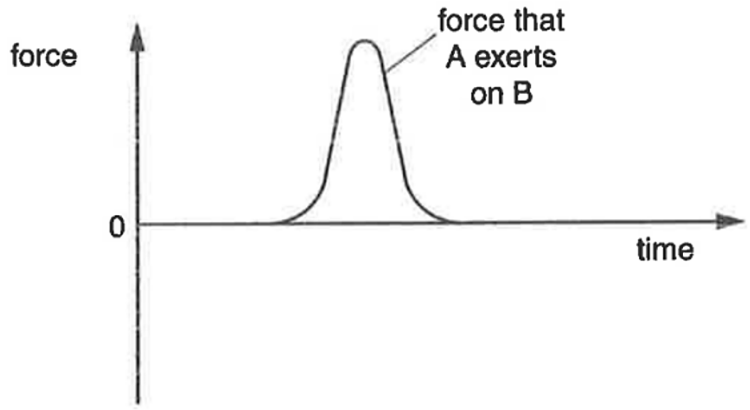


Fig. 2.1

(a) **(i)** Copy Fig. 2.1 and show in Fig. 2.2 a graph of the force that B exerts on A.



Fig. 2.2 [1]

(ii) Explain your answer to **(i)**.

……..

……..

……..

……..

……..

……..

……..…… [1]

(b) In a collision, when a truck of mass 12 000 kg runs into the front of a car of mass 1200 kg, a constant force of 72 000 N acts for 0.25 s.

Calculate the change in velocity of

(i) the car,

change in velocity = ... m s^{-1} [2]

(ii) the truck,

change in velocity = ... m s^{-1} [2]

(c) State and explain how the force on the driver could be reduced by safety features in a typical car.

……..

……..

……..

……..

……..…… [2]

[Total: 8]

2 The Earth has a mass M of 5.97×10^{24} kg. A satellite of mass m orbits the Earth in a circular geostationary orbit at a distance d above the Earth's surface.

(a) Show that the radius of the Earth, R, is approximately 6370 km.

[1]

(b) By considering the orbit of the geostationary satellite, show that the distance d above the Earth's suface is 35900 km.

[2]

(c) For the satellite of mass m that is in a circular orbit of radius r around the Earth of mass M, show that its total mechanical energy E can be expressed as $E = -\frac{GMm}{2r}$.

[2]

(d) The satellite has a mass of 490 kg and carries 360 kg of lithium battery pack with energy density of 324 Wh kg^{-1}, of which 65% of its energy is converted to useful energy for orbital manoeuvres.

(i) Show that the amount of useful energy is 2.73×10^8 J.

[1]

(ii) Calculate the maximum orbital radius r_{max} the satellite can reach from its geostationary orbit.

r_{max} = ... m [2]

[Total: 8]

4 A non-uniform L-shaped beam of weight 10 N is attached to a wall using a hinge. The longer part of the beam is 50 cm and the shorter part of the beam is 10 cm. The weight of the beam acts at a point on the beam that is at a horizonal distance of *x* away from the hinge. The beam is held at rest using a cord, such that the longer part of the beam is horizontal. *AO* is horizontal while *OC* is vertical. The tension in the cord is 5.0 N, at an angle of 60° from *OC*, as shown in Fig. 4.1.

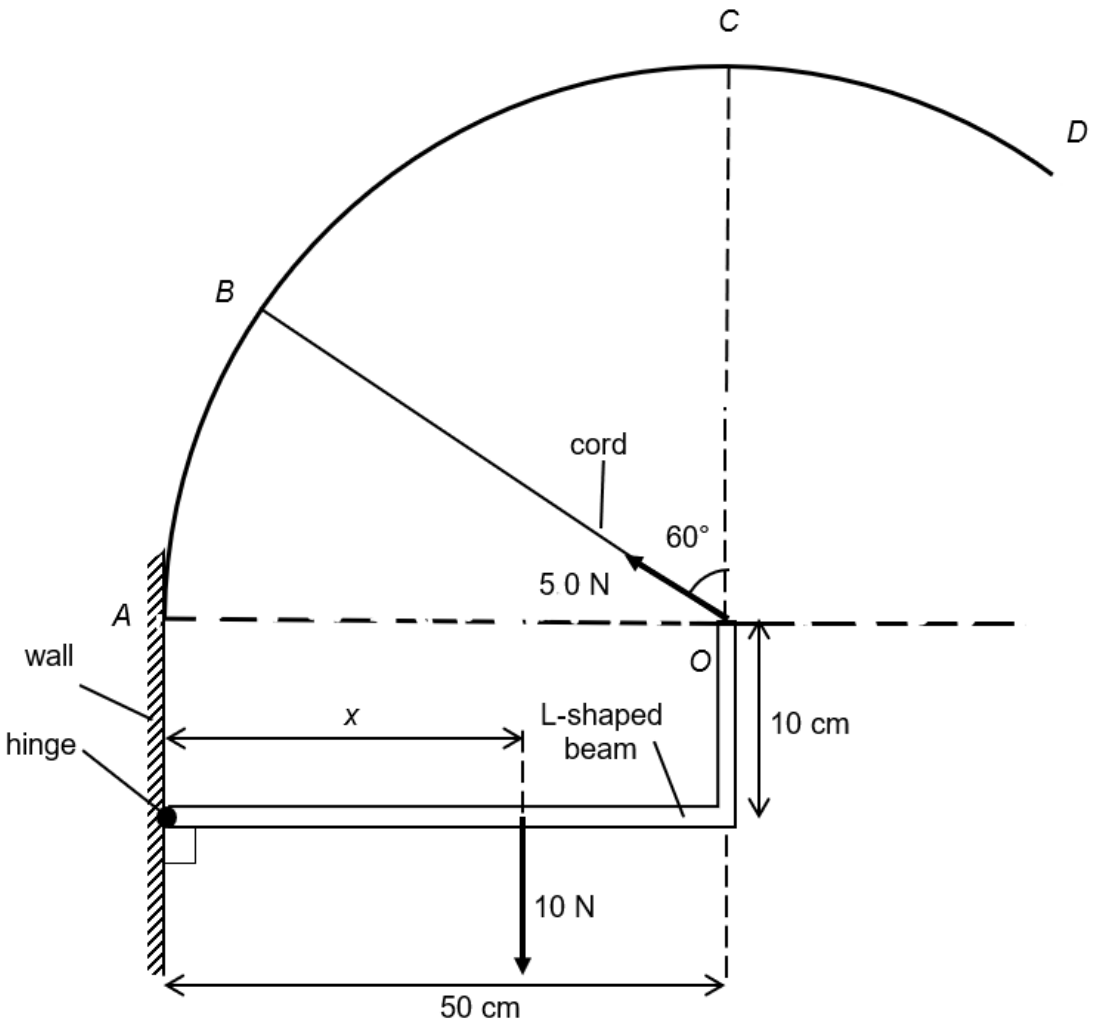


Fig. 4.1

(a) On Fig. 4.1, draw and label the contact force *R* acting on the L-shaped beam due to the hinge.

[2]

(b) By considering the forces acting on the beam, calculate the magnitude of R.

magnitude of R = ... N [3]

(c) Determine the value of x.

x = ... cm [2]

(d) $ABCD$ are different points on a circle where the centre is at point O. C is at the point vertically above point O. The cord moves along the arc of the circle such that the beam remains stationary.

(i) State how the magnitude of the contact force R would change as the cord moves from B to C.

……

……

…… [1]

(ii) State how the direction of the contact force R would change as the cord moves from C to D. Mathematical calculation is not required in this part.

……

……

…… [1]

[Total: 9]

Answer **all** the questions in the spaces provided

1 A ball falls vertically in air from rest. The variation with time t of distance d moved by the ball is shown in Fig. 1.1.

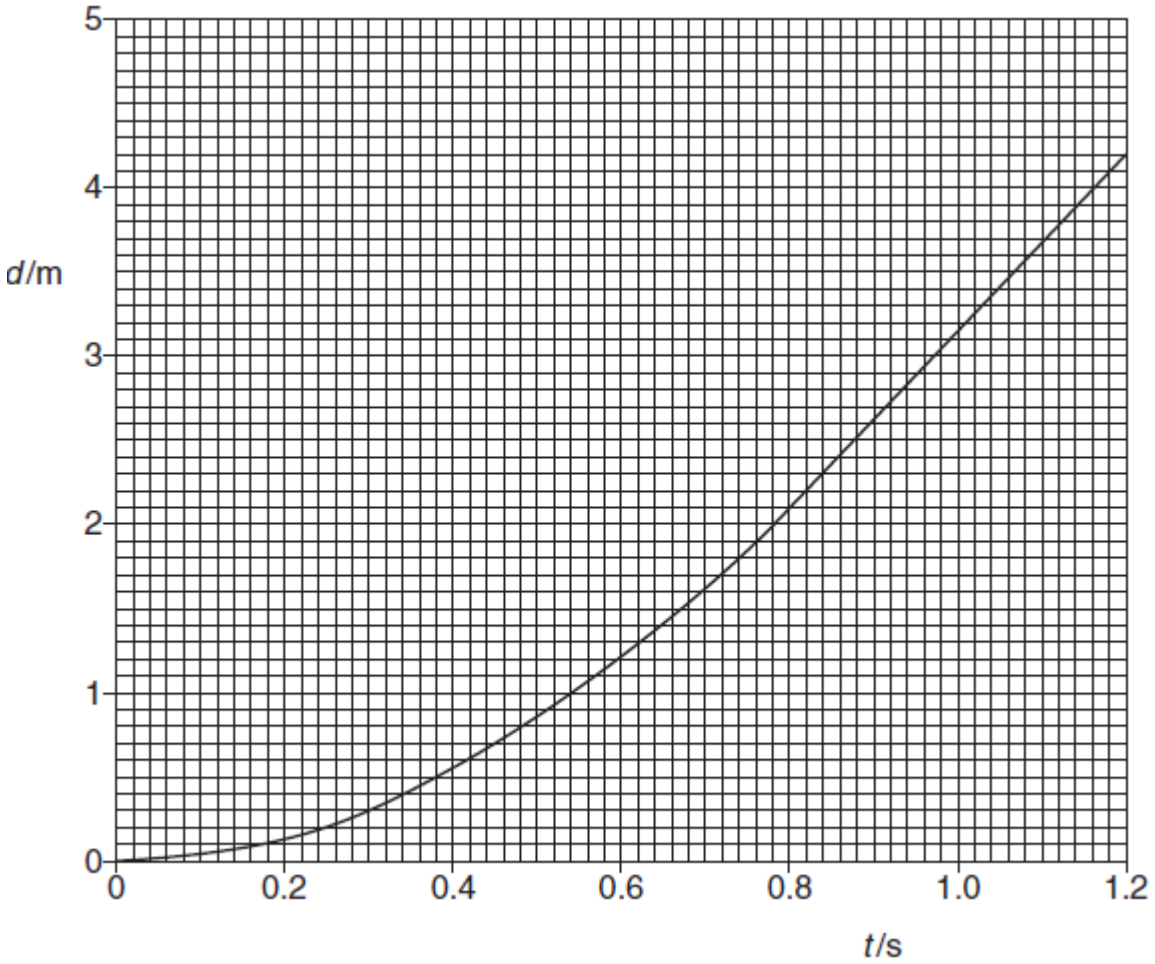


Fig. 1.1

(a) With reference to Fig. 1.1,

1. explain how it can be deduced that the ball is initially at rest,

……

……

…… [1]

2. explain, with reference to forces acting on the ball, why air resistance is not negligible.

……

……

…… [2]

(b) Use Fig. 1.1 to determine the speed of the ball at a time of 0.40 s after it has been released.

speed = ... m s^{-1} [2]

(c) The same ball is now tossed vertically into the air with velocity V to reach a maximum displacement of 2.0 m.

Sketch a graph on the axis below to show the variation with time t of the velocity v of the ball from when it is tossed to when it returns to its original position.

Mark the instant at which the magnitude of acceleration is g = 9.81 m s^{-2} as t_1 and the instant at which the ball returns to original position as t_2. No further calculations are required.

[3]



[Total: 8]

2 **(a)** State Newton's second law of motion.

……

……

…… [1]

(b) At a construction site, sand is allowed to fall vertically from a stationary hopper at a steady rate of 100 g s^{-1} on to a horizontal conveyor belt. The belt is driven by an electric motor and moves at a constant velocity of 5.0 cm s^{-1} as shown in Fig. 2.1.

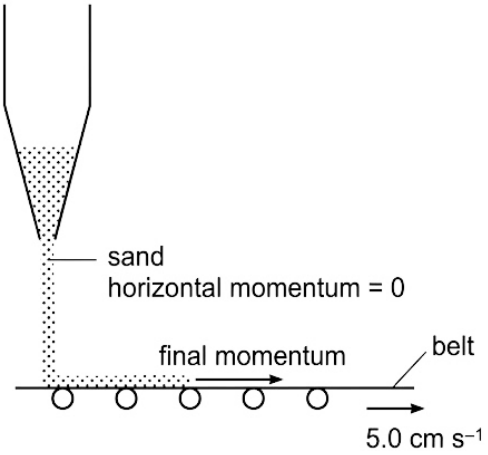


Fig. 2.1.

(i) Using Newton's laws of motion, explain why the sand exerts a horizontal force on the conveyor belt.

……

……

……

…… [2]

(ii) Show that the magnitude of this horizontal component of force exerted on the sand by the belt is 0.0050 N.

[1]

(iii) Determine

1. the rate at which the motor does work, P, to keep the belt moving at constant velocity, and

P = ……..….. W [1]

2. the rate at which the sand gains kinetic energy, K.

K = ……W [1]

(iv) State the type of collision between the sand and the conveyor belt. With reference to your answers in (b)(iii), describe the energy transfers that occur as the sand is brought to the speed of the belt.

…….

…….

…….

…… [2]

[Total: 8]

Section A

Answer all the questions in this section in the spaces provided.

1 **(a)** Define moment of a force.

……

…….. [1]

(b) Fig. 1.1 shows a uniform metal beam pivoted at one end O and held in equilibrium in a horizontal plane by a spring attached to its other end P.

The spring has an unstretched length of 7.0 m and obeys Hooke's law.

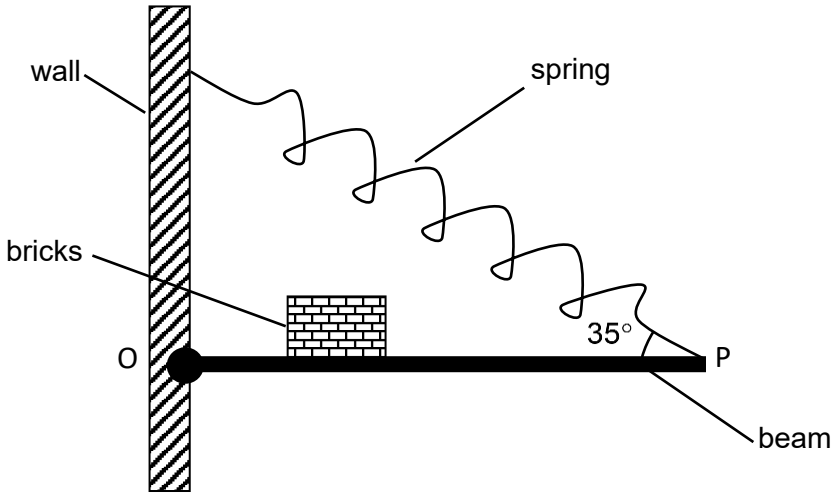


Fig. 1.1 (not to scale)

The beam has mass 190 kg and length 6.8 m. The spring makes an angle of 35° to the beam. A pile of bricks of total mass 120 kg is placed on the beam.

(i) Show that the extension of the spring is 1.3 m. [1]

(ii) On Fig. 1.1, draw labelled arrows to show all the forces acting on the beam. [2]

(iii) Right now, the centre of the pile of bricks is 5.0 m away from end P. Calculate the spring constant k of the spring.

k = …… N m^{-1} [2]

(iv) Calculate the elastic potential energy E_P of the spring.

E_P = …… J [2]

(v) The spring in Fig. 1.1 is now replaced with an inextensible string such that the beam remains in the original horizontal position.

Describe and explain what would happen to the force exerted by the string on the beam as the pile of bricks is slowly moved towards the pivot O while maintaining equilibrium.

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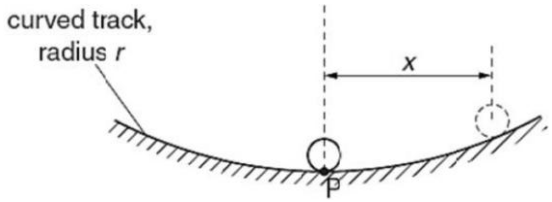
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…… [2]

[Total: 10]

2 A small ball of mass 90 g rests at point P on a smooth curved track of radius r, as shown in Fig. 2.1.



The ball is moved a small distance to one side and is then released. The horizontal displacement x of the ball is related to its horizontal acceleration a towards P, by the expression

$$a = -\frac{gx}{r}$$

(a) Explain why the horizontal oscillation undertaken by the ball can be taken as simple harmonic motion.

……

……

……[2]

(b) The radius r of curvature of the track is 28 cm.

Determine the angular frequency of the oscillation.

angular frequency = …….. rad s^{-1} [2]

(c) After some time, the variation with time t of the displacement x of the ball is plotted as shown in Fig. 2.2. T is the period of oscillation.

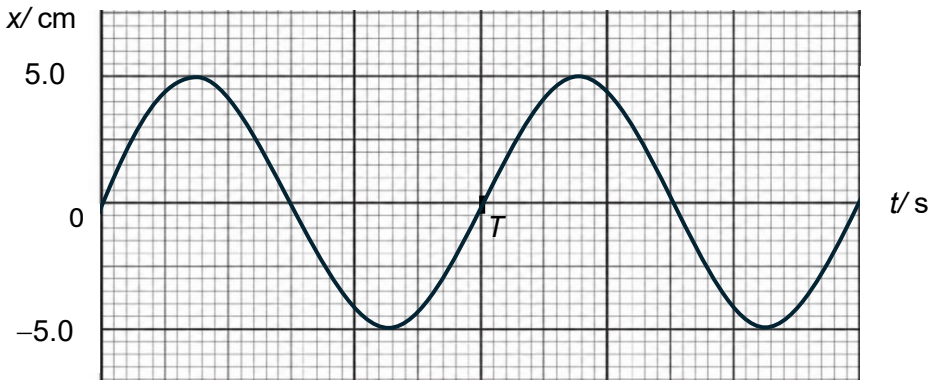


Fig. 2.2

(i) On the axes of Fig. 2.3, sketch the variation with time t of the acceleration a of the ball. [2]

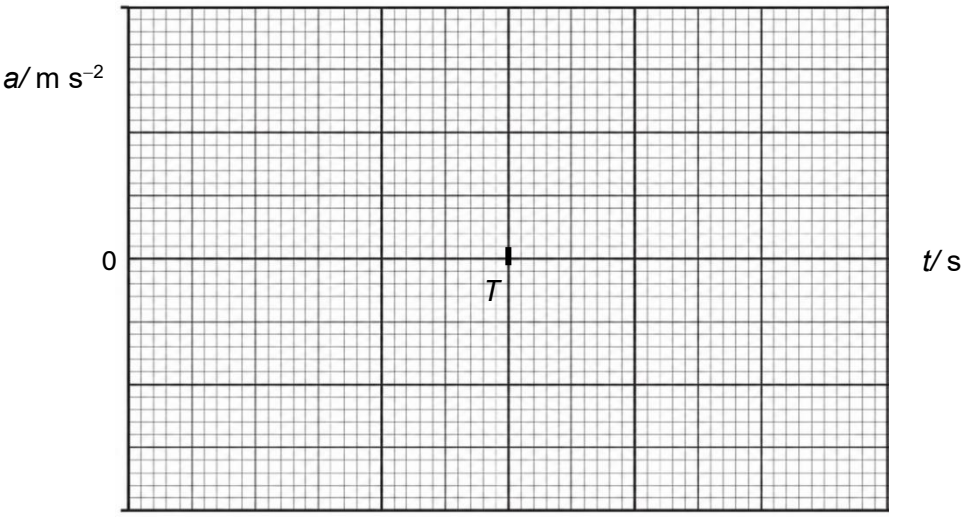


Fig. 2.3

(ii) The track is subsequently brought to a room with high humidity and moisture is formed on the track such that the ball comes to a rest after 10 oscillations.

On the axes of Fig. 2.2, sketch the variation with time t of the displacement x of the ball, for the first two periods after moisture has formed. [2]

(iii) After many oscillations, the amplitude of oscillation decreased from 5.0 cm to 2.5 cm.

Calculate the loss in energy of the oscillation.

loss in energy = …… J [2]

[Total: 10]

5 **(a)** Explain why the gravitational field strength is equal to the acceleration of free fall.

.....

.....

.....

..... [2]

(b) The mean radius of the Earth is 6370 km.

Show that the mass of the Earth is 5.97×10^{24} kg.

[1]

(c) Determine the gravitational potential at the surface of the Earth.

gravitational potential = ... J kg^{-1} [1]

(d) Explain why the gravitational potential is negative.

.....

.....

.....

..... [2]

[Total: 6]

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Answer **all** the questions in the spaces provided.

1 Figure 1.1 shows the variation with the speed v of the force F_D driving a car forward and the resistive force F_R that opposes the motion of the car.

The mass of the car is 1200 kg.

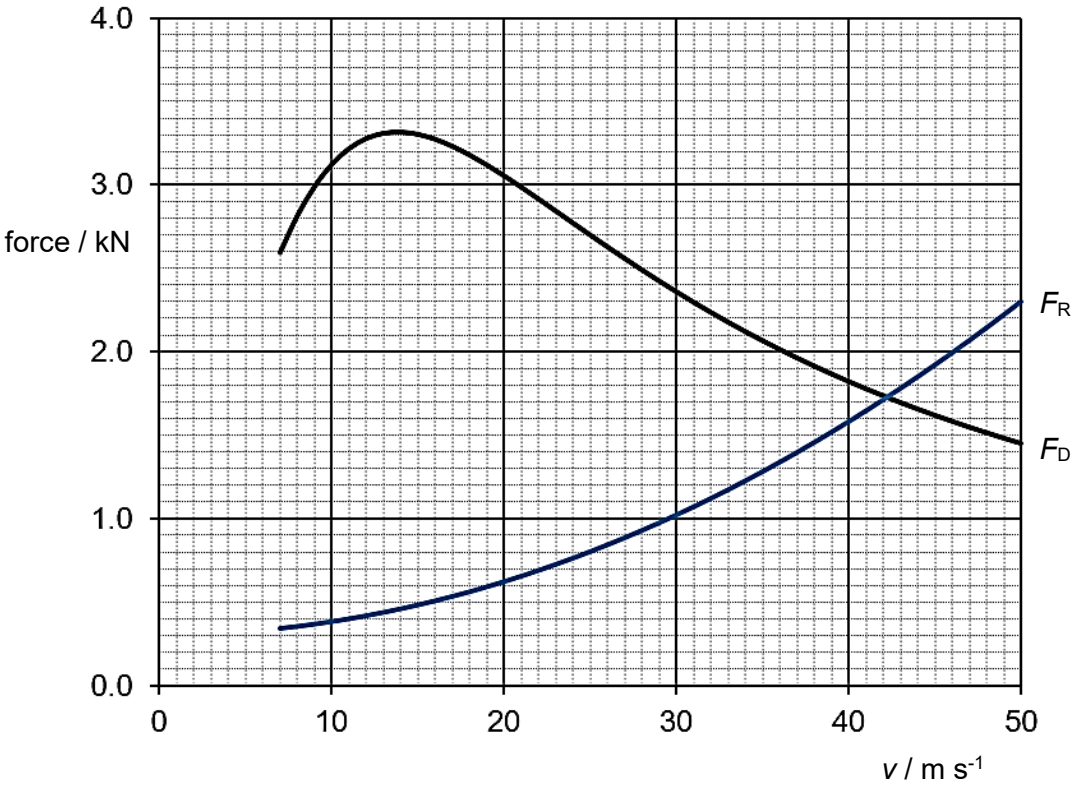


Figure 1.1

(a) Use Figure 1.1 to determine, for the car travelling along a horizontal road at **maximum resultant force,**

(i) the speed at which the car is travelling,

speed = ... m s^{-1} [1]

(ii) its maximum acceleration,

acceleration = ... m s^{-2} [2]

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(iii) the rate of increase of kinetic energy of the car.

rate of increase of kinetic energy = ... W [2]

(b) The car now travels up a slope inclined at an angle to the horizontal.

(i) Explain why, the maximum speed the car can have is less than that when it is travelling on level road.

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..... [2]

(ii) On Figure 1.1, sketch the variation with speed *v* of the total force opposing the motion of the car up the slope.

Label this graph as **S**.

[1]

[Total: 8]

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2 A ball is projected from point A, as shown on Figure 2.1.

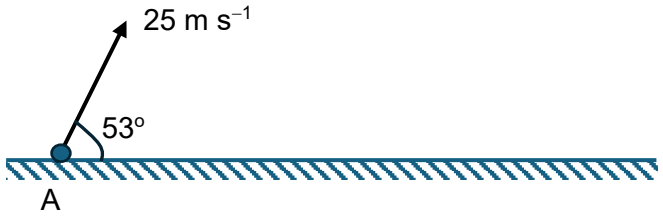


Figure 2.1

The initial velocity of the ball is 25 m s^{-1} at an angle of 53° to the horizontal. The mass of the ball is 0.20 kg.

(a) At a point P on its way to its highest point, the ball's velocity is at an angle of 45° to the horizontal.

(i) Show that the magnitude of the vertical component of the velocity of the ball at point P is 15 m s^{-1}.

[1]

(ii) Calculate the time it takes for the ball to travel from point A to point P.

time = ……..…… s [1]

(b) Calculate the kinetic energy of the ball at a height of 8.0 m.

kinetic energy = …… J [2]

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(c) On Figure 2.2, sketch the variation with height of the kinetic energy of the ball.
No numerical values are needed.

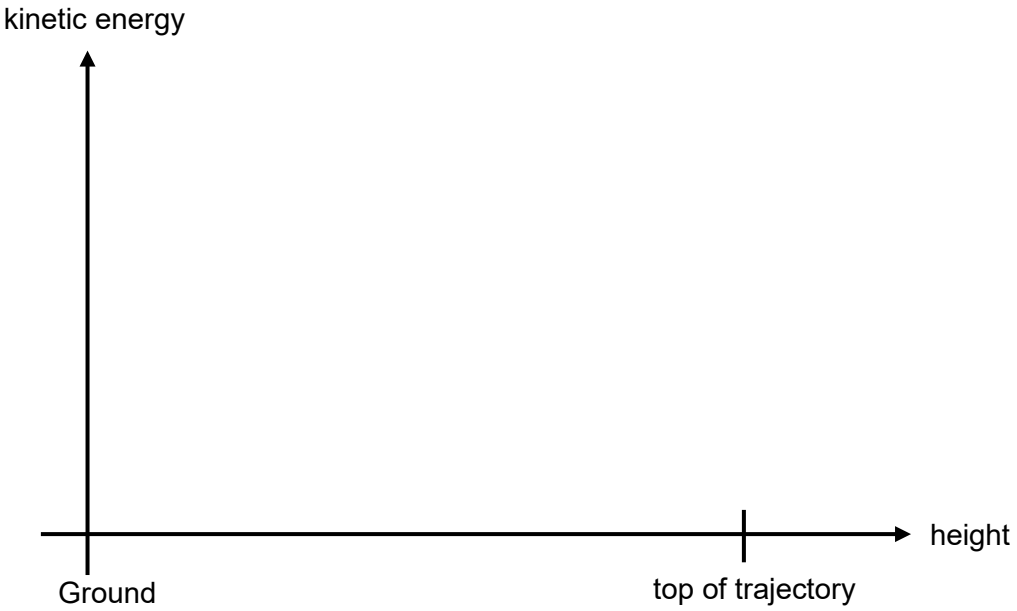


Figure 2.2

[2]

[Total: 6]

9 Read the passage below and answer the questions that follow.

The Sun is the star at the heart of our solar system. Its gravity holds the solar system together, keeping everything – from the biggest planets to the smallest particles of debris – in its orbit. The connection and interactions between the Sun and Earth drive the seasons, ocean currents, weather, climate, radiation belts and auroras.

The Sun has produced a near-constant radiant power of 4.0×10^{26} W for around 5 billion years and is only halfway through its supply of hydrogen fuel. Hydrogen is fused into helium in the Sun's core in a three-stage process known as the proton-proton cycle.

Stage 1 Deuterium Formation**:** Two hydrogen nuclei collide to form a deuterium nucleus releasing a positron and a neutrino

Stage 2 Helium-3 Formation**:** The deuterium nucleus quickly captures another proton, creating a helium-3 nucleus and releasing a gamma-ray photon

Stage 3 Helium-4 Formation: Two helium-3 nuclei collide, resulting in one helium-4 nucleus and releasing protons back into the core

Fusion requires extremely high temperatures to give the hydrogen nuclei (protons) sufficient energy to overcome electrostatic repulsion. But even at the high temperature of 1.5×10^7 °C in the core (Figure 9.1), the average energy of the protons is not enough to allow fusion. This provided a puzzle for physicists which was finally explained by a process called quantum tunnelling.

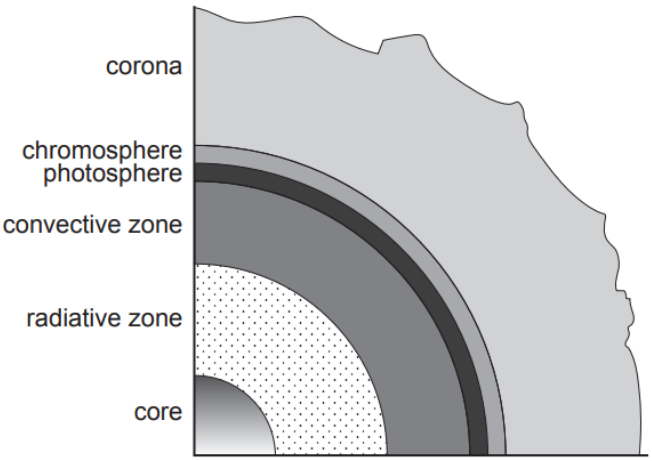


Figure 9.1 Cross section of the Sun (not to scale)

The solar wind is a continuous stream of charged particles (plasma), mainly protons and electrons, escaping the Sun's hot outer atmosphere (corona) at speeds of 500 km s^{-1}.

The Earth's magnetosphere deflects most of the solar wind (Figure 9.2). For particles that sneak through, they spiral in toward the poles, where they can contribute to the glowing lights of the auroras (Figure 9.3) when they smash into particles in the atmosphere.

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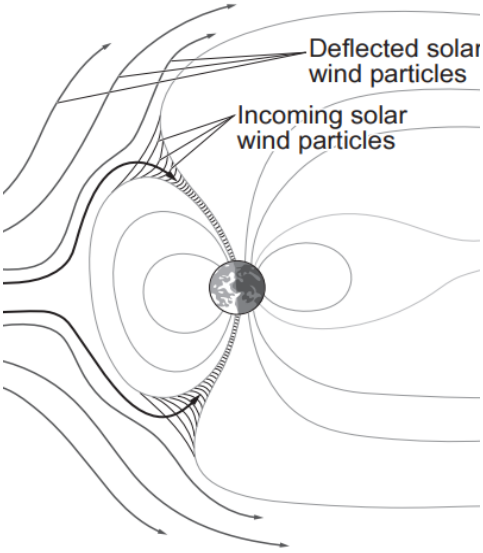


Figure 9.2 Earth's magnetic field deflecting much of the solar wind



Figure 9.3 The aurora borealis seen in Norway

Solar flares on the other hand are associated with coronal mass ejections, rapid release of plasma and ions of a typical total energy of 10^{20} to 10^{25} J. The ejected material travels through space at relativistic speeds and carries its own magnetic field which can temporarily compress Earth's magnetic field. The resulting change in Earth's magnetic field strength can burn out transformer stations in the power grid leading to blackouts. Solar research is hence an important field.

One of the most ambitious missions is the Parker Solar Probe, launched in 2018 to perform a series of passes over the Sun in a gradually changing orbit. The closest approach that the Parker Probe has made to the Sun is 1.4 x 10^{10} m. The power from the Sun at small distances is very large and is expected to raise the temperature of the probe to extremely high values. We can get an idea of the temperature it may reach if we make the simplifying assumption that the probe is in thermal equilibrium – that it radiates as much power per square metre as it absorbs. If this is the case, the probe would be behaving as a "black body radiator" and we can use Stefan's Law to estimate its temperature.

Stefan's Law states that the power absorbed or emitted per unit area by a black body in thermal equilibrium = σT^4, where T is the thermodynamic temperature and σ = 5.7 x 10^{-8} W m^{-2} K^{-4}.

(a) The equation for stage 1 of the proton-proton cycle is shown below:

$$ {}_{1}^{1}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{1}^{2}\mathrm{H} + {}_{+1}^{0}\mathrm{e} + {}_{0}^{0}\nu $$

Write the equations for the subsequent stages:

Stage 2: ……

Stage 3: ……..……..……..…… [2]

(b) (i) Assuming the protons behave like an ideal gas, show that the average kinetic energy of a proton in the core of the Sun is 3.1×10^{-16} J.

[1]

(ii) Two protons need to come within 10^{-14} m of each other to fuse.

Show that on average, the protons do not have enough energy to fuse.

Explain your reasoning clearly.

[3]

(c) (i) Explain what is meant by the wave function in quantum mechanics.

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..... [1]

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(ii) Quantum tunnelling is a quantum mechanical phenomenon in which an object such as an electron or atom passes through a potential energy barrier that, according to classical mechanics, should not be passable due to the object not having sufficient energy to pass or surmount the barrier.

Figure 9.4(a) shows a particle of energy E passing through a potential energy barrier U_0.

Figure 9.4(b) gives the wave function of the particle in the corresponding regions of space.

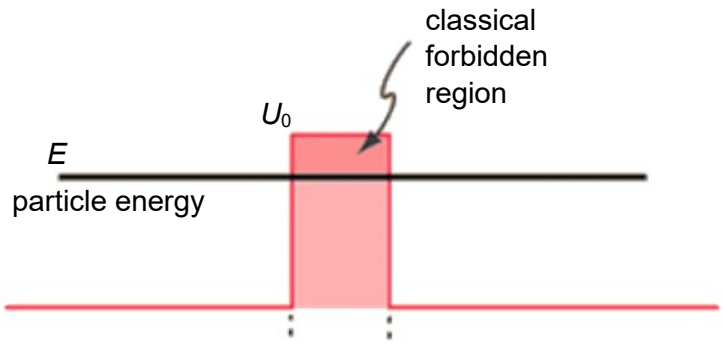


Figure 9.4(a)

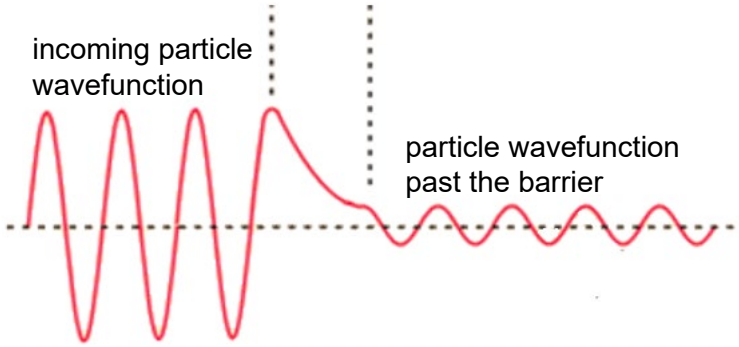


Figure 9.4(b)

With reference to Figure 9.4(b),

1. explain how the amplitude of the wave function shows the particle can exist beyond the potential barrier.

.....

.....

.....

..... [2]

2. by considering the wavelength of the wave function beyond the potential barrier, comment on the energy of the particle after quantum tunnelling.

……

……

……

…… [2]

(d) Consider an electron in the solar wind travelling at an angle of 25° to the direction of the Earth's magnetic field of 40 μT as shown in Figure 9.5.

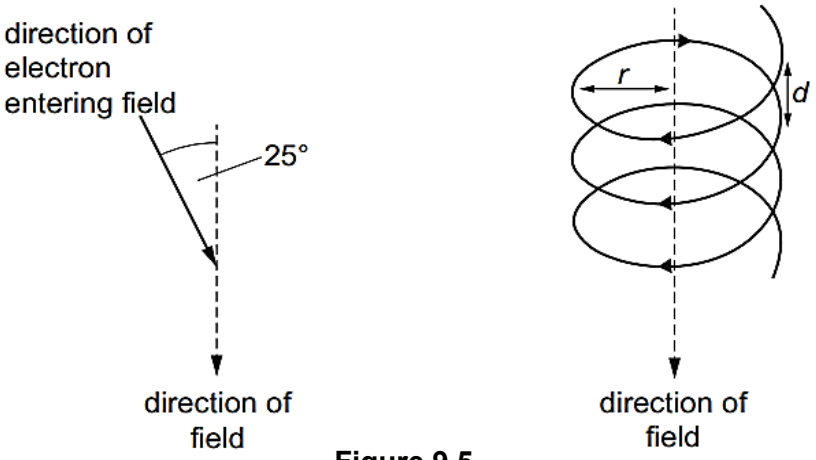


Figure 9.5

(i) By considering the horizontal and vertical components of its velocity, calculate the vertical distance *d* the electron travels between a full rotation around the field line.

distance *d* = ……m [4]

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(ii) State and explain one change to the path if it were a proton instead of an electron coming in at the same angle.

.....

.....

.....

..... [2]

(iii) Explain the mechanisms that cause the atoms in the atmosphere to produce light (aurora) when smashed by the charged particles.

.....

.....

.....

.....

..... [2]

(e) **(i)** Estimate the temperature of the Parker Solar Probe at its closest distance to the Sun. Assume the Sun to be a point source of power.

temperature = ... K [3]

(ii) Suggest why missions investigating the Sun such as the Parker Solar Probe can have an importance beyond scientific discovery.

.....

..... [1]

[Total: 23]

~ End of Paper ~

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Section A

Answer **all** the questions in the spaces provided.

1 A uniform rod AB of weight 12 N is freely hinged to a wall at A. It is held by a force *F* acting from B at an angle of 35° to the vertical as shown in Figure 1.1.

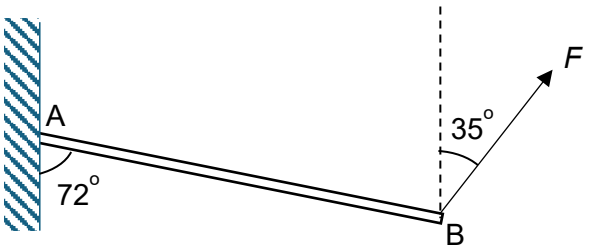


Figure 1.1

(a) Show that the force *F* is 6.0 N.

[1]

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(b) (i) Draw on Figure 1.1 the force R acting on the rod at A.

[1]

(ii) Determine the magnitude of R.

R = ... N [3]

[Total: 5]

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2 This question is about artificial satellites that orbit the Earth. You may treat the Earth as a perfect sphere with the centre of gravity at its centre. All other celestial bodies and their effects need not be considered.

The radius of the Earth is 6370 km and the mass of the Earth is 5.97×10^{24} kg.

(a) (i) Determine the orbital period of a geostationary satellite in seconds.

period = ... s [1]

(ii) Determine the orbital radius of a geostationary satellite to the centre of the Earth.

radius = ... m [2]

(b) The Molniya orbit (Figure 2.1) is a highly elliptical orbital path designed to provide continuous communication coverage to polar regions by positioning its furthest point directly over the North Pole.

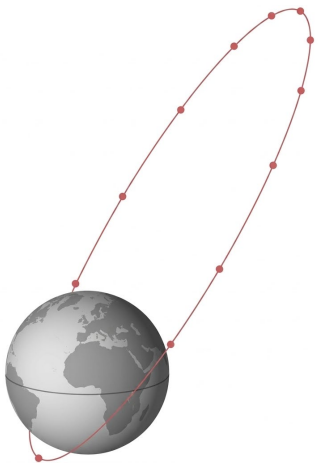


Figure 2.1

A satellite of mass 1500 kg is placed into this orbit. The closest distance the satellite approaches the Earth's centre is 6870 km while its furthest distance from the Earth's centre is 46200 km.

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(i) Calculate the change in gravitational potential experienced by the satellite as the satellite moves from the closest to the furthest points to Earth.

change in gravitational potential = ... J kg^{-1} [2]

(ii) Calculate the change in kinetic energy of the satellite as the satellite moves from the closest to the furthest points to Earth.

change in kinetic energy = ... J [2]

(iii) With reference to your answer in **(b)(ii)**, explain why designing the orbit to have its furthest point over the pole makes it highly effective for continuous communications to the polar region.

.....

.....

..... [1]

[Total: 8]

Section B

Answer **one** question from this section in the spaces provided.

9 **(a)** State the principle of conservation of momentum.

.....

.....

..... [2]

(b) Two frictionless trolleys A and B with attached identical springs approach each other along a horizontal line, as illustrated in Figure 9.1.

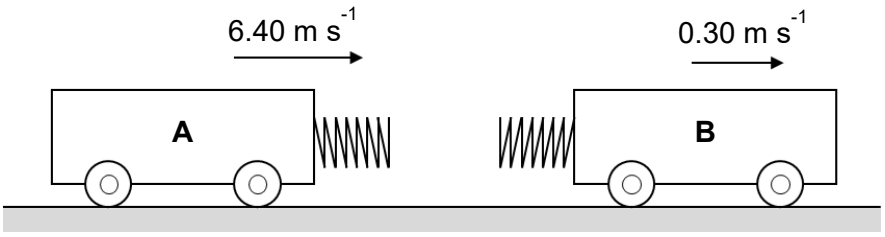


Figure 9.1

Trolley A has mass 1.2 kg and a velocity of 6.40 m s^{-1}. Trolley B has mass 5.2 kg and a velocity of 0.30 m s^{-1}.

The two trolleys collide elastically. After collision, trolley A reverses its direction of motion.

(i) Show that the speed of trolley B after the collision is 2.6 m s^{-1}.

[3]

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(ii) Figure 9.2 shows the velocity-time graph of trolley B.

Sketch also on Figure 9.2, the velocity-time graph for trolley A and label it as **A**.

Graph of velocity against time

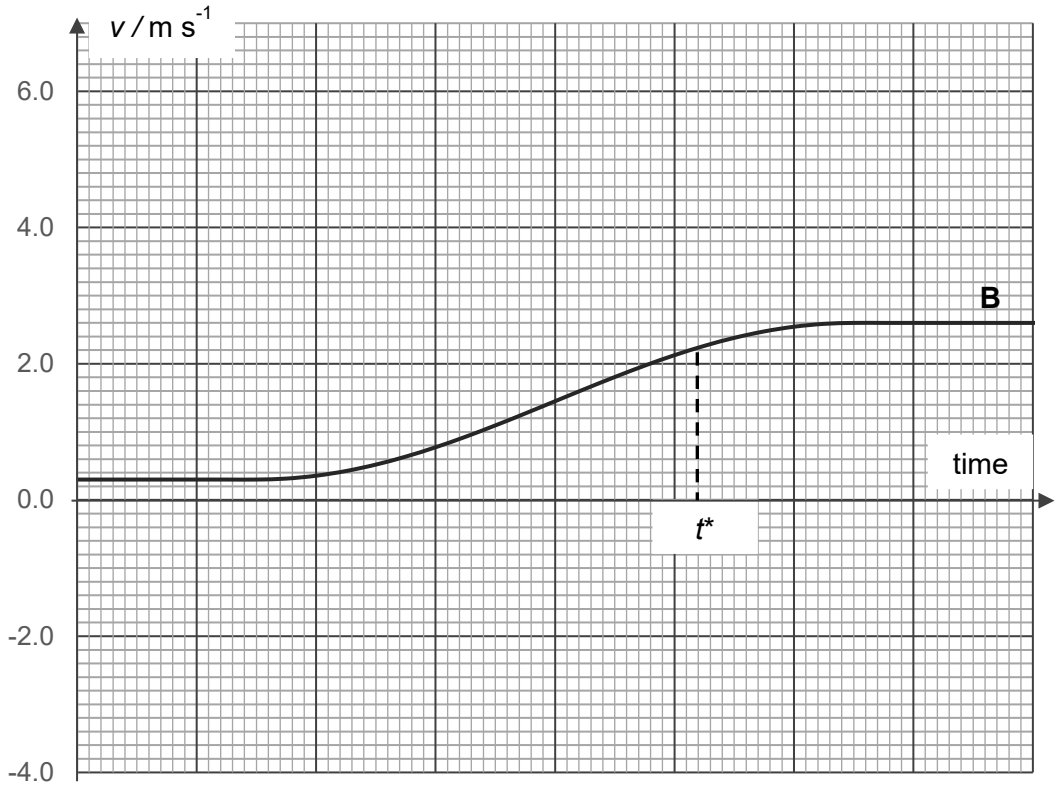


Figure 9.2

[3]

(iii) Explain how the contact force on trolley B by trolley A at time *t** can be determined from Figure 9.2.

.....

.....

..... [2]

(iv) Describe the change in the total kinetic energy of the system of trolleys A and B throughout the collision.

.....

.....

..... [2]

(v) Calculate the maximum elastic potential energy stored in *each* spring.

maximum elastic potential energy = ... J [3]

(c) After the collision of the two trolleys, trolley B moves to the edge of the platform and falls through a vertical height of 10.0 m before being brought to rest in a duration of 0.100 s, without rebounding.

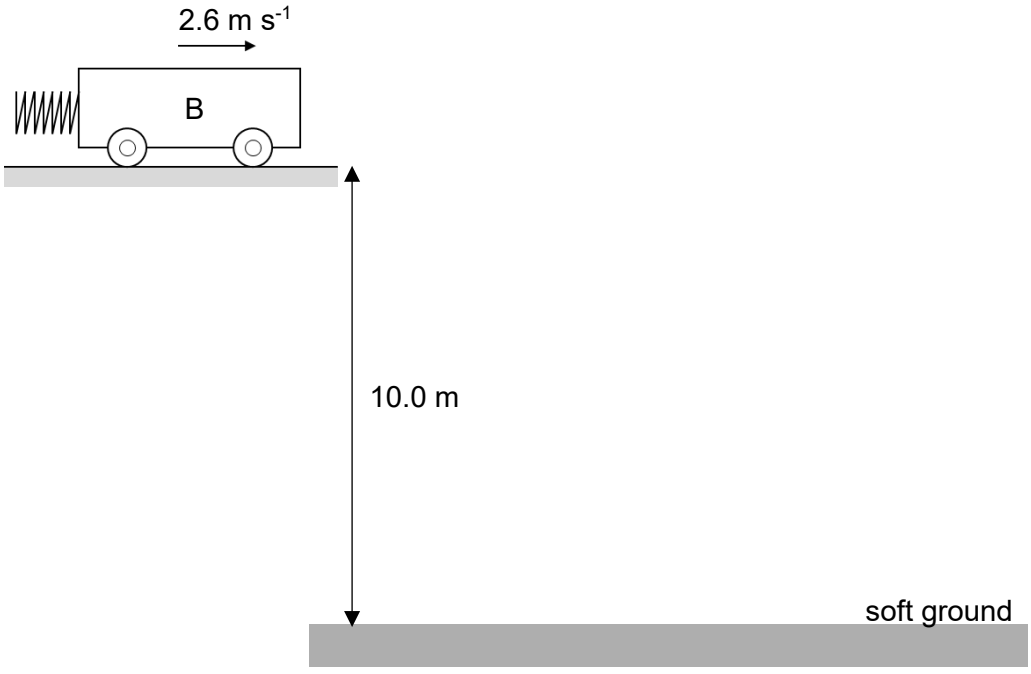


Figure 9.3 (not to scale)

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(i) Determine the average vertical contact force acting on the trolley by the soft ground.

force = …… N [4]

(ii) Explain what happens to the impulse on the trolley if it were to land on hard ground and rebounded instead.

.....

..... [1]

[Total: 20]

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a body to be in equilibrium.

.....

.....

.....

.....[2]

(b) Block A of mass 1.7 kg is connected by a light inextensible string to a counterweight B of mass *M* via a system of two massless and frictionless pulleys P and Q. Block A is balanced on a smooth plane inclined at 45°.

Pulley Q is a double pulley made up of two fixed wheels of different radii attached to a common axle, with the radii of the smaller and larger wheels *r* and 3*r* respectively as shown in Fig. 1.1.

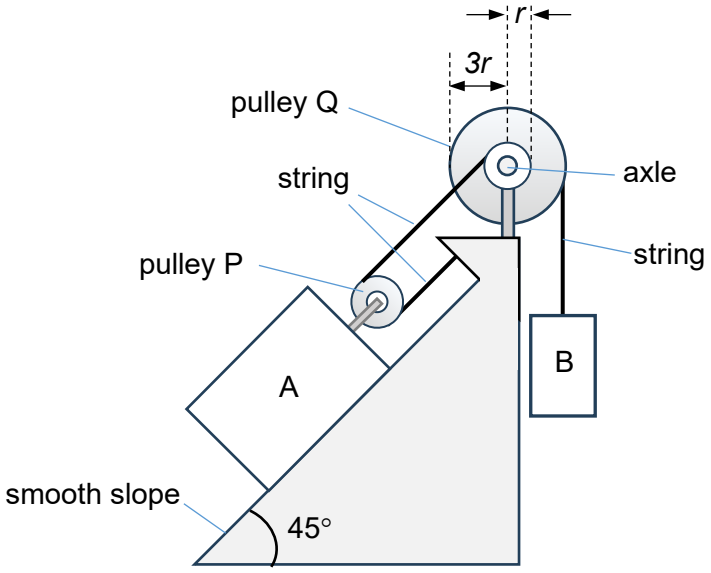


Fig.1.1

(i) Block A is in equilibrium. In Fig. 1.2, draw and label all forces acting on block A.

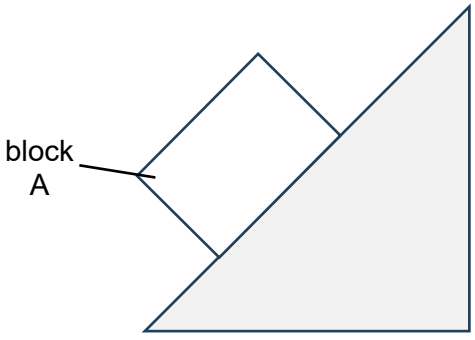


Fig.1.2

[2]

(ii) Calculate the tension in the string connected to pulley P.

tension = ... N [2]

(iii) Determine the mass *M* of counterweight B.

M = ... kg [2]

(iv) Suggest and explain a change that can be made to the pulley such that block A will accelerate up the slope.

.....

.....

.....

.....[2]

[Total: 10]

2 A plane is flying with a velocity of 240 m s^{-1} at an angle of 30° with respect to the horizontal, as shown in Fig. 2.1. At an altitude of 2.0 km, a projectile is released from the plane. The projectile hits the target on the ground. Assume air resistance is negligible.

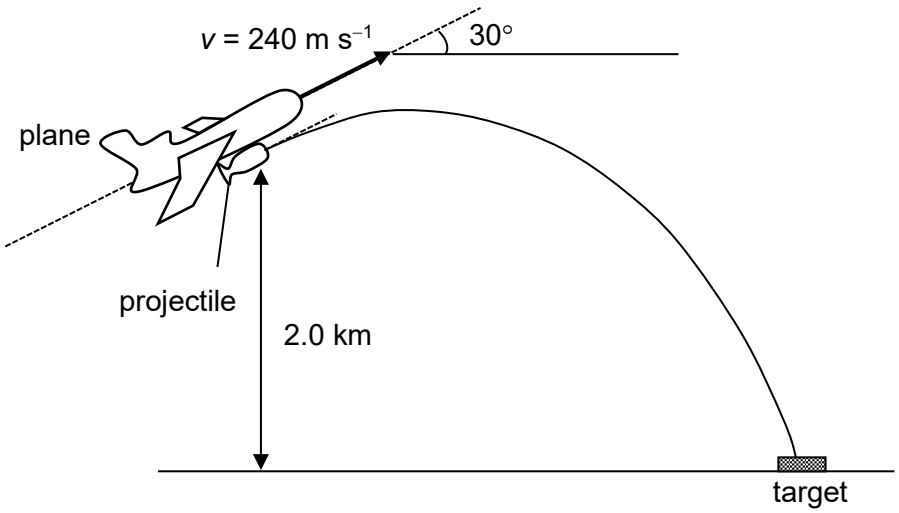


Fig. 2.1

(a) Determine the horizontal displacement of the projectile from the point of release to the target.

horizontal displacement = ... m [3]

(b) Determine the speed of the projectile just before it hits the target.

speed = ... m s^{-1} [2]

(c) (i) Without further calculation, on Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile from the time it was released to the time it hits the target. Label this line **P**. Include values on your axes. [1]

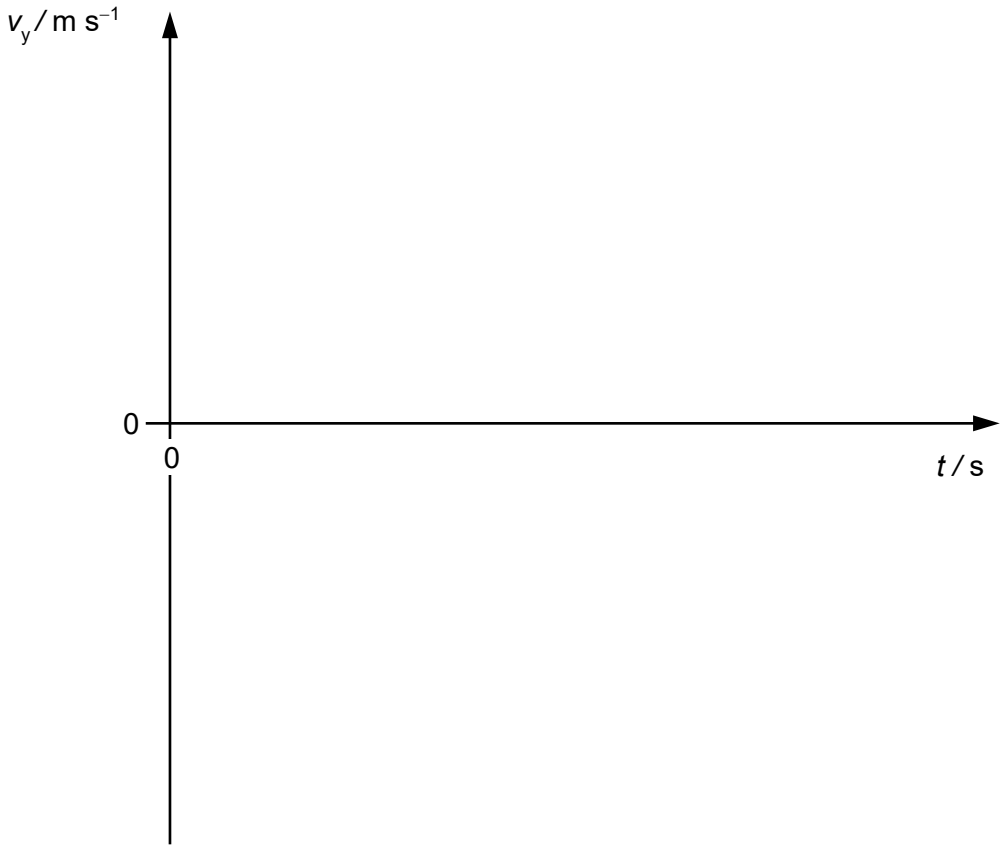


Fig. 2.2

(ii) On Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile if air resistance was **not** negligible. Label this line **Q**. No calculations are required. [2]

[Total: 8]

3 **(a)** PQRS comprises of a rough slope PQR of height 1.1 m, and smooth horizontal surface RS. PQR of total surface length 2.7 m is made up of a straight section PQ and curved section QR as shown in Fig. 3.1.

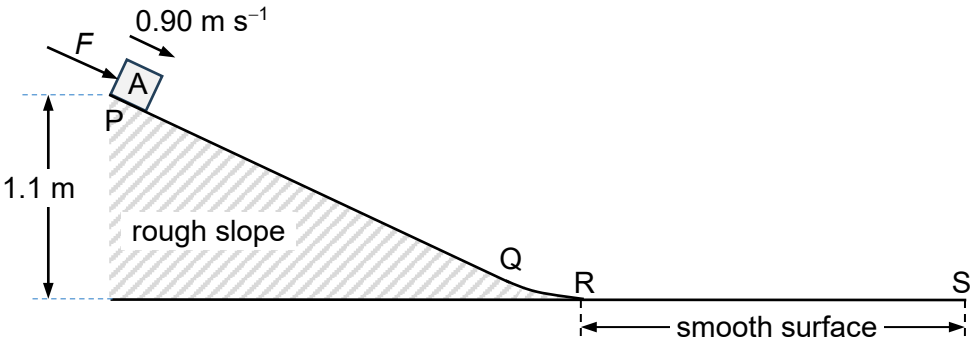


Fig. 3.1

Block A of mass 2.0 kg has an initial speed of 0.90 m s^{-1} at point P on rough slope. A force *F* is applied to block A as it moves along PQR. The magnitude of the *F* is 11 N and its direction is the same as the object's velocity at any instant along surface PQR.

The frictional force between the block A and the surface PQR may be taken to be constant at 6.4 N.

(i) Determine the speed of the block when it reaches R.

speed = ... m s^{-1} [3]

(ii) Use Newton's second law to explain how the speed of block A varies along PQRS.

.....

.....

.....

.....

.....

.....[3]

(b) Block X of mass 0.27 kg moves with a speed 0.67 m s^{-1} on a smooth horizontal surface. Block Y of mass 0.34 kg moves with speed 0.46 m s^{-1} in the opposite direction. A light spring of natural length 13.0 cm and force constant 120 N m^{-1} is attached to block Y as shown in Fig. 3.2.

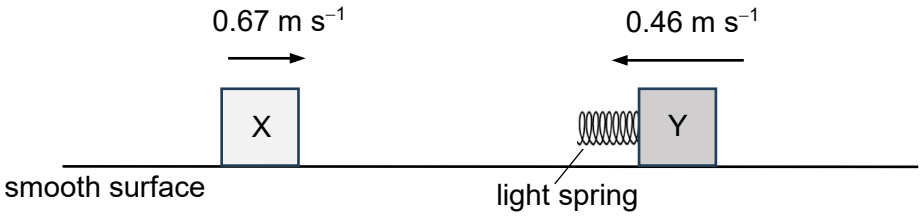


Fig. 3.2

When block X collides elastically head-on with block Y, there will be an instant when the spring experiences maximum compression.

Determine the closest distance of approach of the two blocks.

closest distance of approach = ... m [4]

[Total: 10]

6 Read the passage and answer the questions that follow.

The Juno spacecraft is launched by NASA in 2011, and its mission is to study Jupiter's origin, interior structure, atmosphere, and magnetosphere to understand the solar system's formation by measuring composition, gravity, and magnetic fields with its myriad of sensors on board.

This is also NASA's first mission to Jupiter to use solar panels rather than radioisotope thermoelectric generators (RTGs) for powering the spacecraft. Jupiter orbits the Sun at a much greater distance than Earth, making solar power generation a significant engineering challenge.

Solar radiation spreads out as it travels from the Sun. The solar intensity I at a distance d from the Sun follows the inverse square law. The solar intensity on Jupiter is only 3.72 % that on Earth. To compensate for the low intensity at Jupiter, Juno is equipped with three large solar array wings, as shown in Fig 6.1.

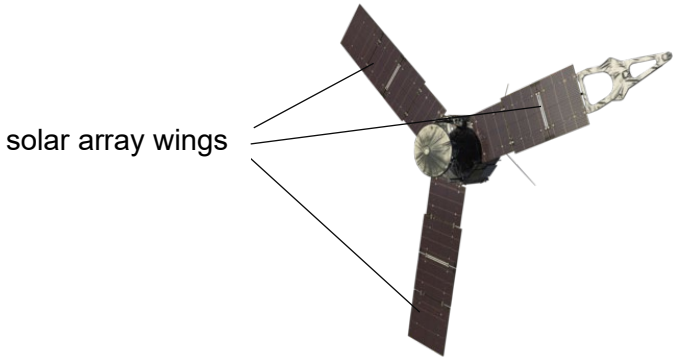


Fig. 6.1

The electrical power P generated by the solar panels on the wings is given by the equation:

$$P = I A \eta$$

where A is the total surface area of the solar panels and η is the efficiency of the solar cells. The efficiency of the cells is affected by the extremely low temperatures in the outer solar system.

Jupiter has an intense radiation belt above its equator that spans a distance of $2R_J$ to $10R_J$ from its centre, where R_J is the radius of Jupiter. Exposure to excessive radiation can cause damage to the spacecraft and its sensors. To minimize such exposure, Juno follows a highly elliptical polar (pole-to-pole) orbit, as shown in Fig 6.2.

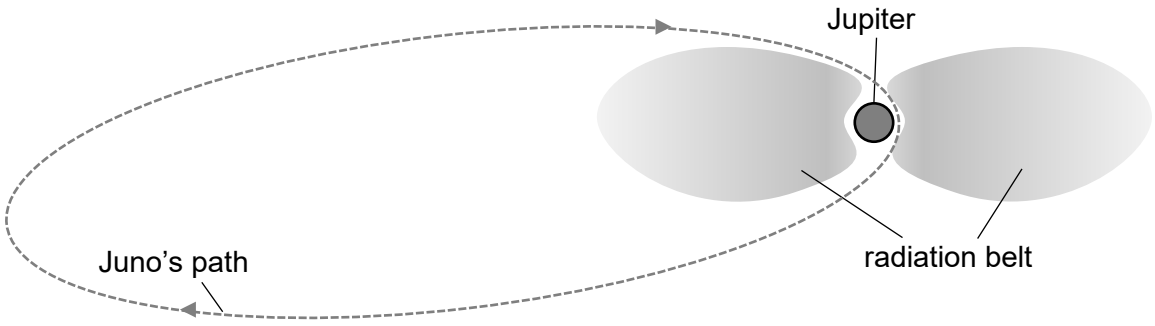


Fig. 6.2

Once Juno enters Jupiter's orbit, its propulsion system is turned off. It is subsequently used sparingly only to correct its path when necessary.

The distance of Juno from the centre of Jupiter in the first three months of 2017 is shown in Fig. 6.3.

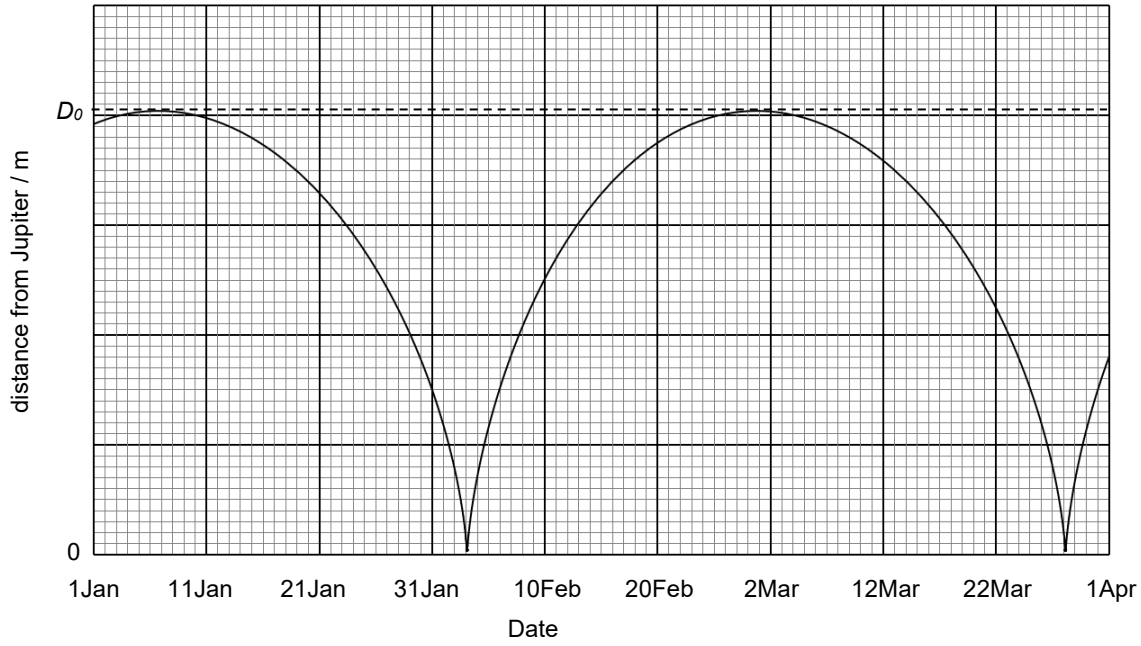


Fig. 6.3

Due to the vast distances involved, it took almost 5 years for Juno to enter Jupiter's orbit from Earth. Communication between Juno and Earth which is conducted via radio waves also experiences significant "light time" delay for signals.

Additional data for Earth, Jupiter and the Juno mission are provided in Table 6.4.

Property	**Value**
Mass of Jupiter M_J	1.90×10^{27} kg
Radius of Jupiter R_J	7.15×10^{7} m
Mean distance* of Earth from the Sun d_E	1.50×10^{11} m
Solar intensity at Earth's distance I_E	1360 W m^{-2}
Total surface area of Juno's solar panels A	60 m^2
Efficiency of solar cells at Jupiter η	24%
Mass of Juno spacecraft m	3625 kg
Perijove (closest distance* to Jupiter) r_p	1.06 R_J
Apojove (furthest distance* from Jupiter) r_a	113 R_J

*center to center

Table 6.4

(a) Explain why the solar intensity I follows an inverse square law as the distance d from the Sun increases.

.....

.....

.....[2]

(b) Using the relation in **(a)** and the data provided, show that the mean distance of Jupiter from the Sun is 7.78×10^{11} m.

[2]

(c) Calculate the total electrical power output generated by Juno's solar panels when Jupiter is at its mean distance from the Sun.

electrical power = ... W [1]

(d) Suggest **one** reason, other than distance from the Sun, why the electrical power output might fluctuate during the mission.

.....

.....[1]

(e) Determine from Fig. 6.3 the period of Juno's elliptical orbit.

period = ... days [1]

(f) Determine the value of D_0 in Fig. 6.3.

D_0 = ... m [1]

(g) Explain why the sum of Juno's kinetic energy and gravitational potential energy remains constant when it is in orbit.

.....

.....

.....[1]

(h) Calculate the difference between the maximum and minimum values for the kinetic energy of Juno during an orbit.

difference = ... J [3]

(i) On Fig. 6.2, indicate point X at which Juno is moving with maximum speed and point Y at which it is moving with minimum speed. [1]

(j) Explain using Fig. 6.2 and your answer in **(i)** how the highly elliptical orbit minimizes the spacecraft's exposure to Jupiter's intense radiation belt.

.....

.....

.....

.....[2]

(k) For a body in a circular orbit, gravitational field strength g is always equal to $\frac{v^2}{r}$, where v is its linear speed and r is the radius of the orbit. In Juno's case, assume g is equal to $\frac{v^2}{r}$ at its perijove and its apojove. Use this relationship to calculate Juno's minimum speed.

minimum speed = ... m s^{-1} [3]

(l) Explain why the relationship $g = \frac{v^2}{r}$ does not hold in general for elliptical orbits.

.....

.....

.....[2]

(m) Using the mean distances of Jupiter and of Earth from the Sun, estimate a minimum value for the "light time" delay for signals to travel from Jupiter to Earth.

delay = ... s [2]

[Total: 22]

End of Paper

Section A

Answer **all** the questions in the spaces provided.

1 Two spheres A and B of masses 3.7 kg and 2.7 kg respectively are attached to two light inextensible strings L_1 and L_2 each of length 0.50 m.

The combination of masses is made to swing in a vertical circle with both strings always in a straight line, as shown in Fig. 1.1.

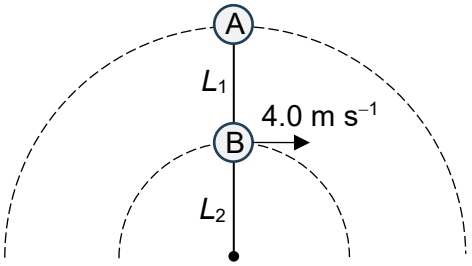


Fig 1.1

At the top of its motion, mass B is travelling at 4.0 m s^{-1}.

(a) Determine the speed of mass A at this instant.

speed of mass A = ... m s^{-1} [2]

(b) Show that the tension in L_1 at this instant is 200 N**.**

[1]

(c) Hence or otherwise, determine the tension in L_2 at this instant.

tension in L_2 = ... N [2]

(d) At another instant in time, the combination of masses is at the bottom of its motion as shown in Fig. 1.2.

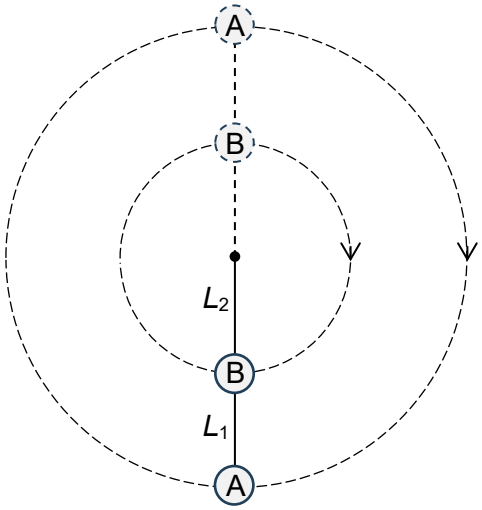


Fig 1.2

A student suggests that because the tensions in strings L_1 and L_2 change continuously during the motion, the total mechanical energy (i.e. kinetic and potential energy) of the system is not conserved.

With reference to tensions in the strings, state whether you agree with this statement and explain your reasoning.

.....

.....

.....

.....[2]

[Total: 7]

2 **(a)** Define gravitational potential at a point

.....

.....

.....[2]

(b) **(i)** The dwarf planet Ceres may be considered to be an isolated uniform sphere of mass 9.4×10^{20} kg and radius 4.7×10^{5} m.

A particle is launched vertically upwards from the surface of the planet. Assuming negligible resistance from the planet's atmosphere, determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the planet Ceres.

speed = ... m s^{-1} [3]

(ii) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is 3.3×10^{-27} kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 240 K.

r.m.s. speed = ... m s^{-1} [2]

(c) The surface of planet Ceres reaches temperatures of approximately 240 K when in direct sunlight.

Use your answers in **(b)(i)** and **(b)(ii)** to suggest a reason why Ceres does not have an atmosphere consisting of hydrogen.

.....

.....

.....[1]

[Total: 8]

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

.....

.....

.....

.....[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

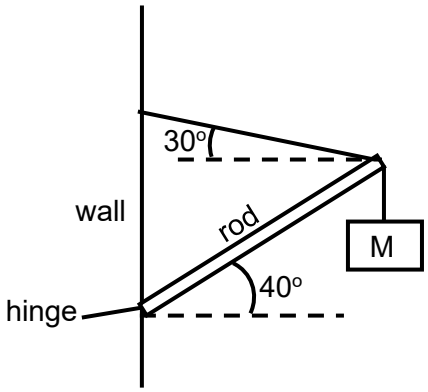


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]

2 A speedboat of mass 450 kg is equipped with two identical motors. Each motor can produce 75 kW of power. When both motors are turned on, its maximum speed is 30 m s^{-1}. Assume that the drag force on the speedboat is proportional to the square of its speed.

(a) Determine the maximum speed of the boat when only one motor is turned on.

maximum speed = ... m s^{-1} [3]

(b) When both motors are turned on, the speedboat takes 9.0 s to reach the maximum speed of 30 m s^{-1}. Determine the amount of energy that is lost to the surroundings during that time.

amount of energy lost = ... J [2]

(c) Sketch, on Fig. 2.1, the variation with time t of the speed of the speedboat v, until it reaches maximum speed.



Fig. 2.1

[1]

(d) When the speedboat is travelling at a speed of 20 m s^{-1}, both motors are turned off at the same time. The speedboat continues to travel an additional distance of 600 m before coming to rest. Determine the average resistive force acting on the speedboat.

average resistive force = ... N [2]

[Total: 8]

3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius *r*, as shown in Fig. 3.1.

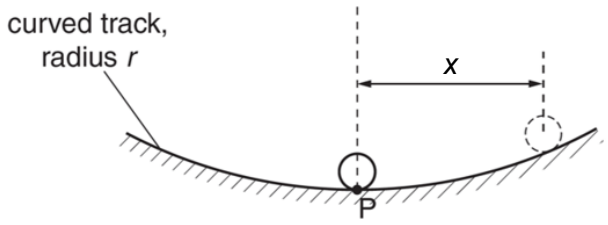


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

.....

.....

.....

.....[2]

(ii) The radius *r* of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

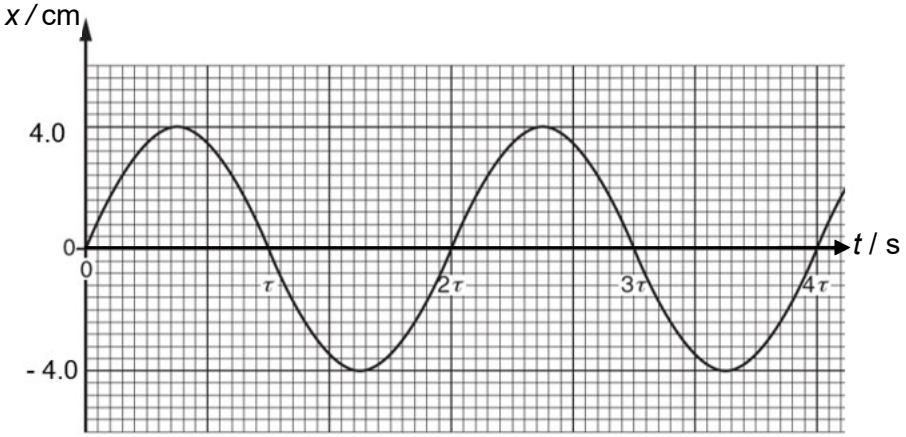


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

maximum kinetic energy= .....J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

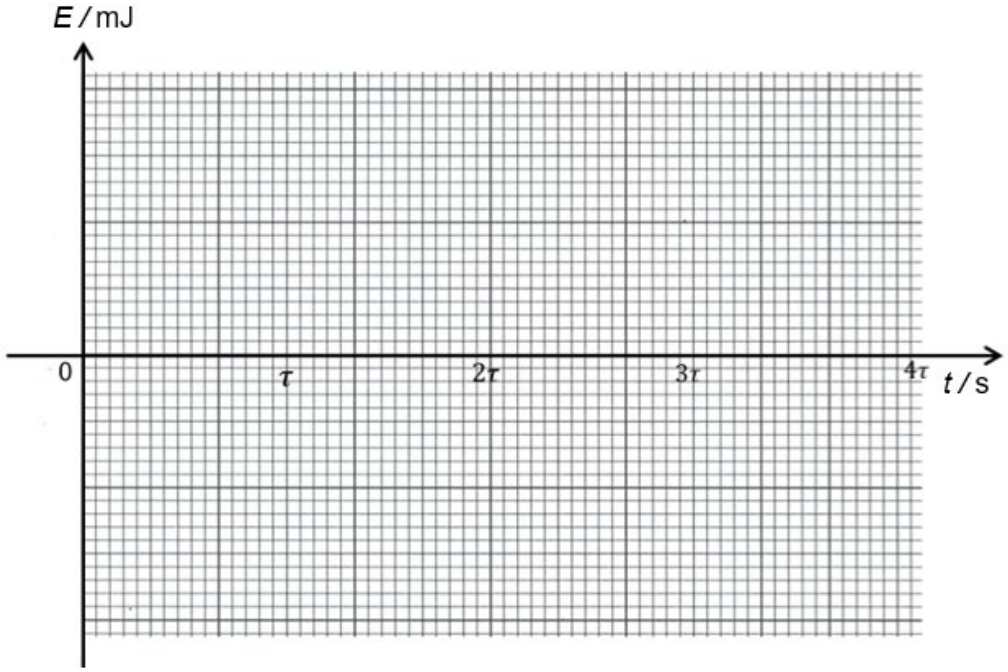


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

[Total: 9]

7 Read the passage below and answer the questions that follow.

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator. It first started up on 10 September 2008 and is the crown jewel within the accelerator complex of the European Council for Nuclear Research (CERN).

Spanning across the Switzerland-France border, the LHC is a large horizontal ring in which high-energy particle beams, most commonly proton beams, are accelerated to speeds close to the speed of light in opposite directions. These proton beams circulate around the ring more than 11000 times a second as they are made to collide multiple times at different interaction points where the beams cross each other. By studying the fundamental particles produced in the aftermath of these proton-proton collisions, scientists hope to gain new insights and answers to some of the most important unsolved mysteries in Physics.

Inside the accelerator, the protons travel around the ring for hours along two separate beam pipes, both of which are kept at an ultrahigh vacuum. As the protons circulate around the LHC, they are repeatedly accelerated by strong electric fields along a short linear accelerator within the ring.

The protons are guided around the accelerator ring along the beam pipes by strong magnetic fields generated by thousands of electromagnets of different types and sizes, all of which need to be constantly kept at extremely low temperatures of about 1.9 K. This is why the LHC also houses the world's largest cryogenic system, using about 1.0×10^7 kg of liquid nitrogen and 1.3×10^5 kg of liquid helium to maintain the operation of the magnets.

The most common type of these electromagnets is called dipole magnets, which help bend the protons' trajectories so that they travel successfully along the curved beam pipes. These dipole magnets consist of many coils of electric cables placed next to and at many different sections along the beam pipe, each carrying a large electric current. The dipole magnets are contained and fastened together with strong non-magnetic stainless-steel collars.

Fig. 7.1 shows the cross-section of one of the dipole magnet setups at a section of a beam pipe of the LHC. The large number of electric cables are grouped into two double-layered regions around the beam pipe, labelled as region 1 and region 2 in Fig. 7.1. These electric cables are arranged parallel to the beam pipe such that when a current is travelling along these cables, the magnetic flux density produced by these cables at the centre of the beam pipe helps bend the protons' trajectories so that they can travel along the curved beam pipe.

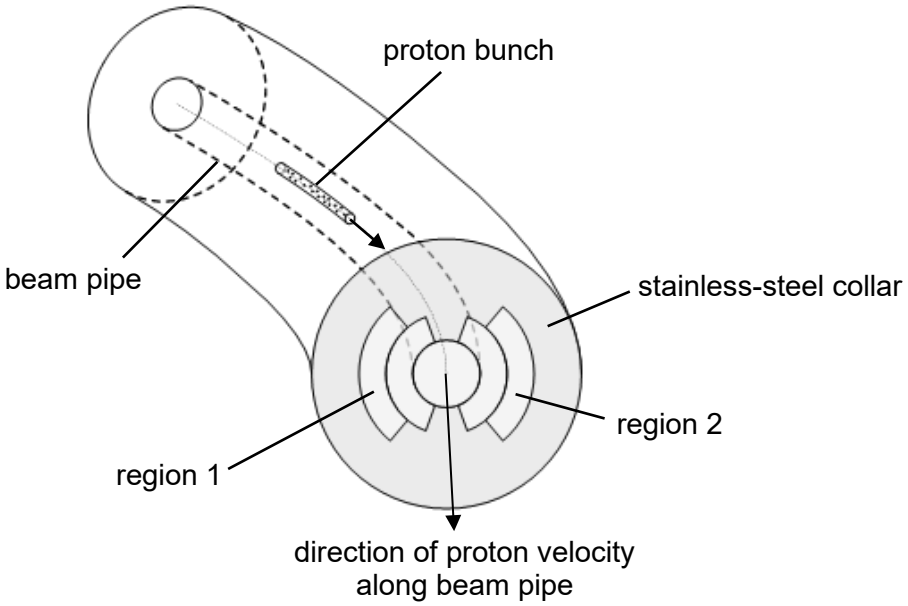


Fig. 7.1
(The electric cables parallel to the beam pipe are not shown for clarity.)

An important parameter in determining the functionality of a particle accelerator like the LHC is the beam current, which is defined as the average rate of flow of charge in the beam pipe. However, during normal operation, the protons do not travel in the beam pipe as a continuous beam. Instead, in each proton beam, the protons travel in groups called "bunches", which are approximately cylindrical in shape as shown in Fig. 7.1.

The LHC is built underground with a mean depth of 100 m for various reasons, including minimising the damage caused to the landscape and environment on the surface. In addition, protons travelling around the curved beam pipes accelerate to produce synchrotron radiation in the form of ultraviolet and X-ray photons, and interactions between the protons and the nuclei of any atoms around the beams in the beam pipes also produce ionising radiation. The Earth's crust therefore also provides natural shielding from these sources of radiation for anyone living on the surface near the LHC.

Table 7.1 shows some important data and parameters of the LHC under normal operating conditions.

Table 7.1

total circumference of LHC ring	26659 m
diameter of beam pipe	56 mm
proton energy	21.2 GeV
current in each cable of a dipole magnet setup	11850 A
length of a dipole magnet setup	14.3 m
peak dipole magnetic flux density at centre of beam pipe	8.33 T
beam current	0.58 A
number of bunches per proton beam	2808
length of one bunch	7.48 cm
cross-sectional area of one bunch	1.0 mm^2

(a) Suggest why the inside of the beam pipe needs to be maintained at an ultrahigh vacuum while the LHC is being operated.

.....

.....

.....[1]

(b) State and explain if an accelerator like the LHC can be used to study neutron-neutron collisions.

.....

.....

.....[1]

(c) (i) The protons in the LHC beam pipe travel at a speed of 0.999*c*, where *c* is the speed of light. Show that the period of revolution of the protons around the LHC ring is 8.89×10^{-5} s.

[1]

(ii) By considering the beam current, show that the total charge in the ring is 5.16×10^{-5} C.

[1]

(iii) Use your answer in **(c)(ii)** to determine the number of protons in each bunch within the proton beam.

number of protons = ... [2]

(d) Fig. 7.2 shows a close-up of the cross-section of the dipole magnet setup around the beam pipe in Fig. 7.1, where regions 1 and 2 are represented by combined cables 1 and 2 respectively. Each combined cable carries electric current of the same magnitude.

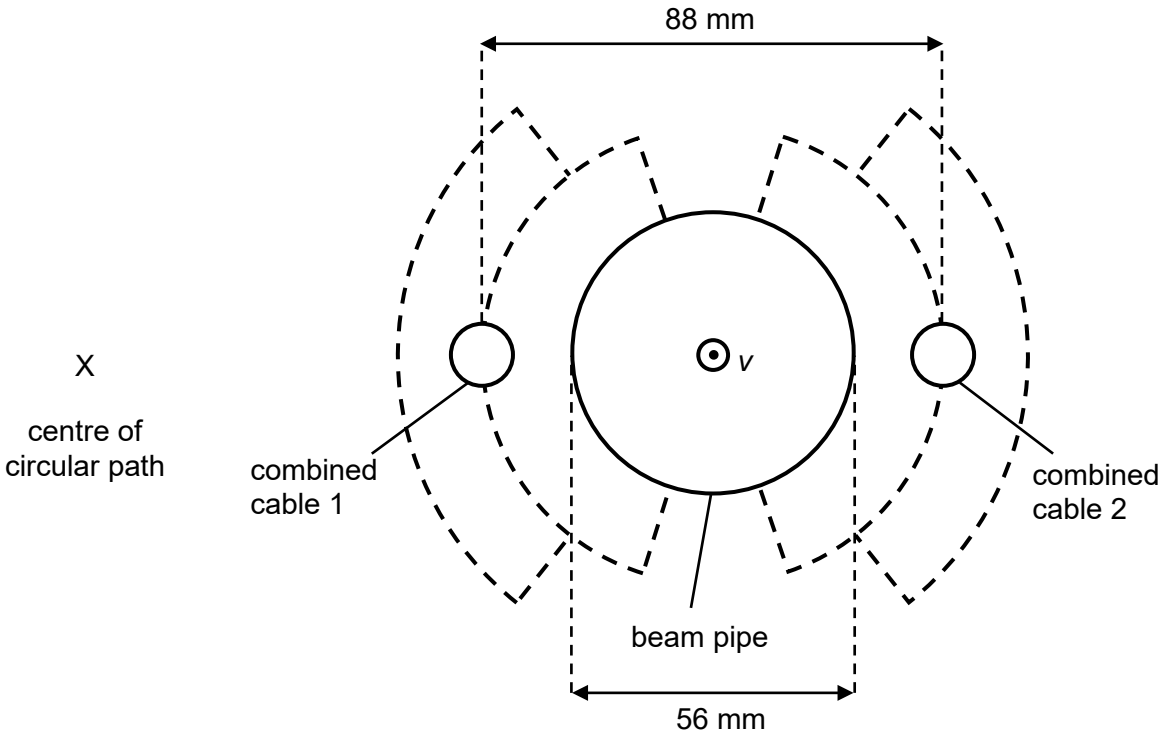


Fig. 7.2

In this model of the dipole magnet setup, the distance between the combined cables is 88 mm, while the diameter of the beam pipe is 56 mm.

The proton velocity *v* in the middle of the beam pipe is pointing out of the page, as shown in Fig. 7.2. The centre of the circular path of the proton's trajectory is on the side of combined cable 1.

(i) On Fig. 7.2, using appropriate symbols and labels, indicate the direction of

1. the magnetic flux density *B* at the centre of the beam pipe,

2. the currents I_1 and I_2 in the combined cables 1 and 2 respectively. [2]

(ii) 1. Determine the current required in each combined cable to produce the peak dipole magnetic flux density at the centre of the beam pipe.

current = ... A [2]

2. Hence, determine the number of cables in each region.

number of cables = ... [1]

3. Suggest a reason why the dipole magnets need to be contained and fastened together with collars that are strong.

Support your reason with appropriate calculations.

.....

.....

.....[3]

(e) Besides dipole magnets, another type of electromagnet used at the LHC are quadrupole magnets.

Quadrupole magnets help “squeeze” the travelling protons within each bunch closer together so that the protons stay travelling along the central axis of the beam pipe in a tightly focused beam. They do so by accounting for the electrostatic repulsion between the protons within each bunch.

(i) By approximating the average volume occupied by a proton in each bunch to be the volume of a cube, show that the average distance between two adjacent protons in a bunch is 8.66×10^{-7} m.

[2]

(ii) Hence determine the magnitude of the average repulsive force between two adjacent protons in a bunch.

average repulsive force = ... N [1]

(iii) The quadrupole magnets also help to keep the protons in the proton beam on their intended path by accounting for the protons falling downwards due to their weight.

Determine the number of rounds a proton will be able to travel around the LHC accelerator ring before falling to the bottom of the beam pipe due to gravity, if there were no quadrupole magnets.

Assume that the acceleration due to gravity in the pipe is 9.81 m s^{-2}.

number of rounds = ... [3]

[Total: 20]

Section A

Answer **all** the questions in the spaces provided.

1 A ball is thrown horizontally from the top of a building, as shown in Fig. 1.1.

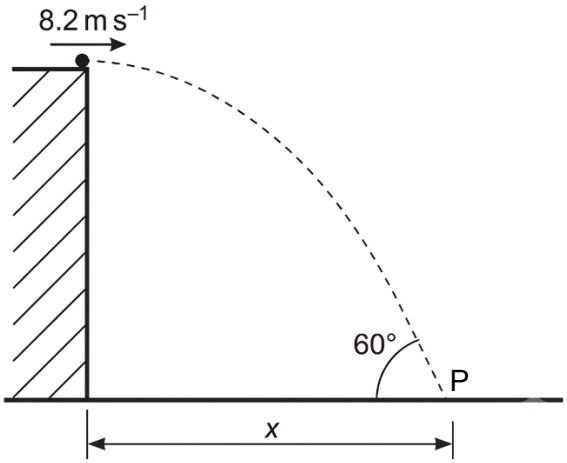


Fig. 1.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The side of the building is vertical. The ball hits the ground at point P at a distance *x* from the building and at an angle of 60° to the horizontal. Air resistance is negligible.

(a) For the ball at point P,

(i) show that the vertical component of its velocity is 14 m s^{-1},

[2]

(ii) determine the vertical distance it has fallen,

vertical distance = ... m [2]

(iii) determine the horizontal distance x.

distance x = ... m [2]

(b) State and explain the effect on your answer to the horizontal distance x if air resistance is not negligible.

.....

.....

.....[2]

[Total: 8]

2 **(a)** State Newton's second law of motion.

.....

.....

.....[1]

(b) A ball of mass 150 g travelling horizontally at a speed of 20 m s^{-1} collides with a vertical wall. Fig. 2.1 shows how the force of impact on the wall, *F*, varies with time *t*.

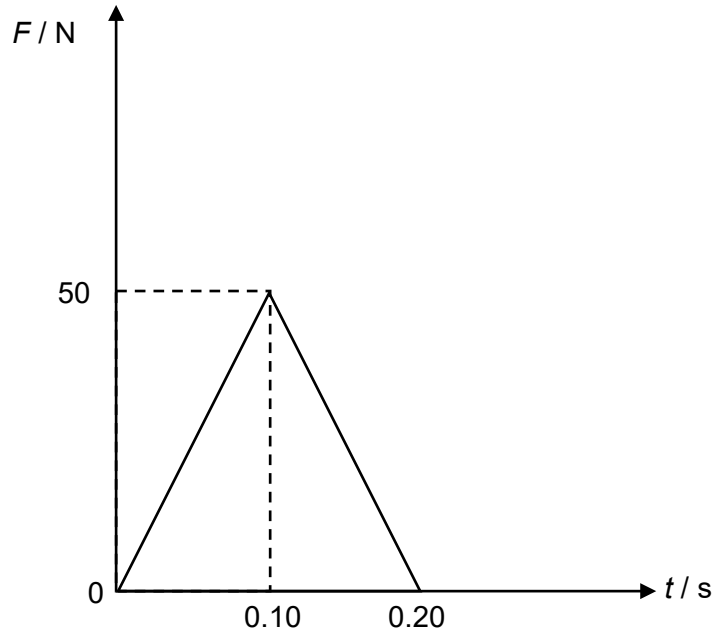


Fig. 2.1

(i) Determine the change in momentum of the ball.

change in momentum = ... N s [2]

(ii) Hence determine the final speed of the ball.

final speed = ... m s^{-1} [1]

(iii) Balls which are identical to the one above are thrown at the wall at the rate of 30 balls per minute. Determine the average impact force on the wall.

average impact force = ... N [2]

(iv) A ball which is equal in mass as the one above is made of a harder material. It also collides with the wall at the same speed as before. Draw on Fig. 2.1 a second graph to show how the impact force on the wall varies with time when this ball collides with the wall. [1]

[Total: 7]

Section B

Answer **one** question from this section in the spaces provided.

8 A theme-park ride consists of two cages. They are moving in a circular path at constant speed v about a horizontal axis. Fig. 8.1 shows the ride at one instant when cage A is vertically above cage B.

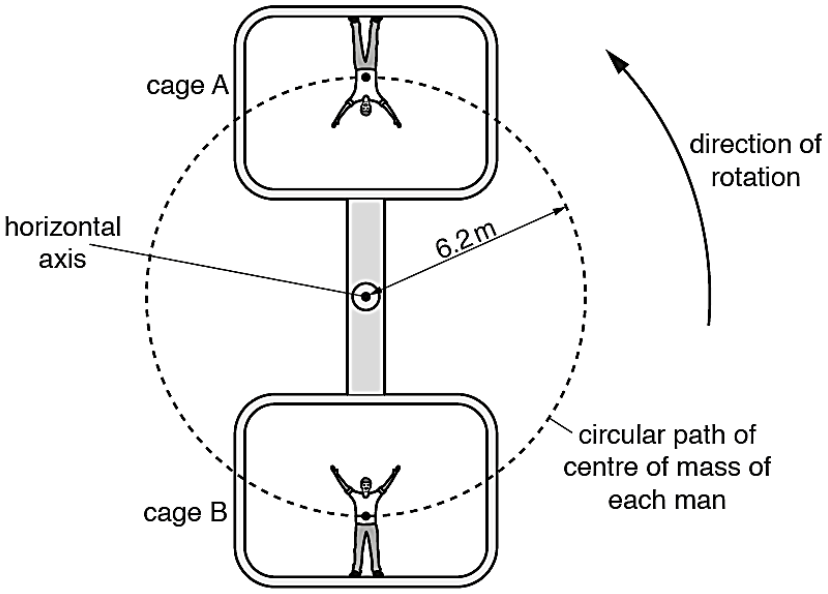


Fig. 8.1

(a) A man is riding in each cage. The mass of each man is 75 kg. The centre of mass of each man is 6.2 m from the horizontal axis. The period of one rotation is 4.1 s.

(i) Determine the speed v of the centre of mass of each man.

speed v = ... m s^{-1} [2]

(ii) Calculate the magnitude of the acceleration of the centre of mass of each man.

acceleration = ... m s^{-2} [2]

(b) Fig. 8.2 shows the forces acting on the man in cage B at the instant the cages are in the positions shown. It shows the man in cage A at that same instant.

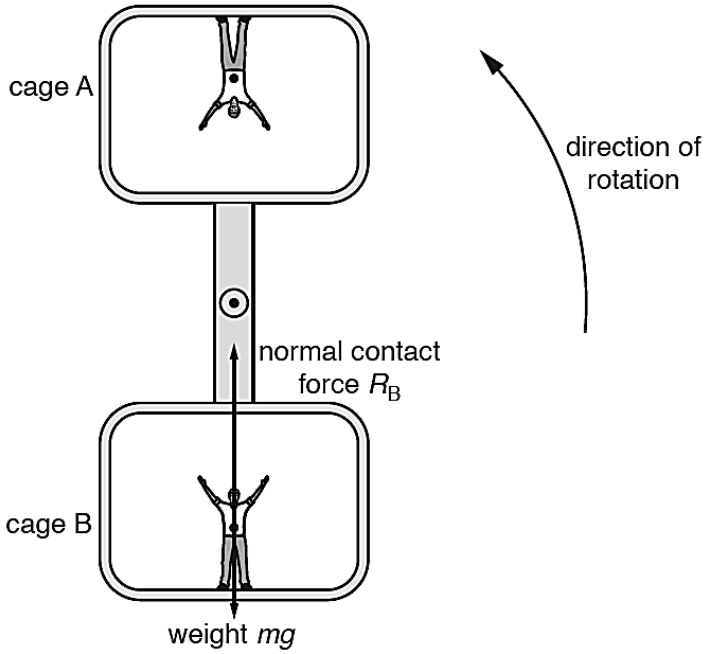


Fig. 8.2

(i) On Fig. 8.2, draw labelled arrows to represent the magnitude and direction of the forces acting on the man in cage A. [2]

(ii) Calculate the magnitude of the normal contact force R_A on the man in cage A at this instant.

R_A = ... N [2]

(c) **(i)** Explain why a minimum value for the speed is needed for the man in cage A to maintain in contact with the floor of his cage.

.....

.....

.....[2]

(ii) Calculate this minimum speed.

minimum speed = ... m s^{-1} [1]

(d) (i) State Newton's Law of Gravitation.

.....

.....

.....[2]

(ii) The astronomer Johannes Kepler showed that the period T of rotation of a planet about the Sun is related to its mean distance R from the centre of the Sun by the expression

$$\frac{R^3}{T^2} = k$$

where k is a constant.

Show that, for planets in circular orbits about the Sun of mass M, the constant k is given by

$$k = \frac{GM}{4\pi^2}$$

where G is the gravitation constant.

[3]

(iii) Earth can be assumed to be in a circular orbit about the Sun with an orbital radius of 1.50×10^8 km.

Use the expression in **d(ii)** to calculate a value for the mass of the Sun.

mass of Sun = ... kg [2]

(iv) Explain why, although planets experience gravitational forces from the Sun, they do not fall into the Sun.

.....

.....

.....

.....[2]

[Total: 20]